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АНГЛИЙСКИЙ ЯЗЫК ДЛЯ СПЕЦИАЛИСТОВ В СФЕРЕ ИНФОРМАЦИОННЫХ ТЕХНОЛОГИЙ. ЧАСТЬ 4

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Хрестоматия состоит из неадаптированных аутентичных текстов, отобранных по принципу профессиональной релевантности. Тексты сохранены в оригинальной стилистике и терминологической сложности, что позволяет обучающимся развивать навыки аналитического чтения, терминологической работы и извлечения информации из первичных источников. Хрестоматия может использоваться для самостоятельной работы студентов, так как содержит вопросы для самопроверки понимания прочитанных текстов.

Для студентов 1-го курса, обучающихся по направлению 01.03.02 «Прикладная математика и информатика» (профиль «Интеллектуальный анализ данных и поддержка принятия решений»).

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How the Internet of Things Works

after Bernadette Johnson

The Internet of Things (IoT), also sometimes referred to as the Internet of Everything (IoE), consists of all the web-enabled devices that collect, send and act on data they acquire from their surrounding environments using embedded sensors, processors and communication hardware. These devices, often called "connected" or "smart" devices, can sometimes talk to other related devices, a process called **machine-to-machine** (M2M) communication, and act on the information they get from one another. Humans can interact with the gadgets to set them up, give them instructions or access the data, but the devices do most of the work on their own without human intervention.

We can monitor our homes and families remotely to keep them safe. Businesses can improve processes to increase productivity and reduce material waste and unforeseen downtime. Sensors in city infrastructure can help reduce road congestion and warn us when infrastructure is in danger of crumbling. Gadgets out in the open can monitor for changing environmental conditions and warn us of impending disasters. Sensors may now collect such data as temperature, moisture, light, chemical levels, heart rate and body movement, location (through accelerometer, gyroscope, GPS chips) etc.

IoT devices often send data to cloud-computing servers where it's then aggregated and analyzed. We can usually access the results via apps or browsers on our mobile devices or home computers. Some can even be set up to update your status on various social networks. Some can bypass your home network entirely via cellular communication.

By some accounts, there are already between 15 and 25 billion connected devices, with the count expected to grow to anywhere from 50 to 212 billion by 2020.

Smart devices can work in conjunction with tagging technology including RFID tags, QR codes, barcodes and the like to obtain data about items. The devices also need a power source, which can include a connection to a power outlet, a solar panel, or even replaceable or rechargeable batteries. Companies are also working on wireless power for a possible future power source.

Some Technical Issues and Solutions

At the moment, a lot of connected devices can talk to the Internet and to our phones, and maybe even some related products, but most of them can't talk to one another because of proprietary hardware and software with differing standards, languages and communication protocols.

There aren't any universal standards or platforms to allow seamless interaction between all smart gadgets and enable you to control them from a central app, but several groups are working on creating standard protocols and software to make interoperability of multiple devices from different manufacturers a reality. The AllSeen Alliance, created by Qualcomm and joined by other companies, is working on an open-source, platform-independent software framework called AllJoyn. Cisco, Samsung, Intel and others are also working on their own open-source platform called IoTivity.

There are a number of other smart device platforms out or coming out, including Apple's HomeKit, Google's Project Brillo, SmartThings, Evrythng, Samsung Artik and Wink. Some of these are hardware hubs and software, and others are just software applications or platforms (either to be set up by a user or implemented by the manufacturers themselves). Some require licensing, and others are open-source. For the most part, they are compatible with multiple types and brands of devices, but none are all encompassing.

Another major issue for the Internet of Things is a finite number of IP addresses. The standard identifier used to route Internet traffic to and from networked devices is the IP (Internet protocol) address. In its first and still-prevalent form (the IPv4 32-bit standard created in 1981), an IP address consists of four numbers separated by periods, each between zero and 255 (that's 256

possibilities for each of the four slots). Because of those limitations, the maximum number of available addresses on this standard is capped at around 4.295 billion and will soon be exhausted. Obvious decision – moving to the new IPv6 system.
IPv6, a 128-bit standard, could allow for more than 340 undecillion addresses (that's 340 followed by 36 zeros). Its format consists of eight sets of four-character hexadecimal values separated by colons. The available number of addresses will still far exceed the number of devices that will be available in the years to come (or maybe ever). An interim solution that a lot of entities already use is network address translation (NAT). NAT allows you to map whole networks of devices to a single IP address, which allows the Internet to see the network as its destination device. Then the network servers differentiate between the devices within that network to get data to and from the right places. This scheme is better for computing equipment within organizations than for gadgets in peoples' homes, however.

Active vocabulary

1. **web-enabled devices** – устройства, подключенные к интернет
2. **embedded sensors** - встроенные сенсоры
3. **unforeseen downtime** - непредвиденный простой
4. **danger of crumbling** – опасность разрушения
5. **hub** – сетевой концентратор
6. **impending disaster** – надвигающаяся катастрофа
7. **cloud-computing servers** – серверы облачных вычислений
8. **to bypass home network via cellular** – обойти домашнюю сеть через сотовую
9. **tagging technologies** – технологии нанесения меток
10. **RFID tag** (от Radio Frequency Identification) - радиометка, электронная метка
11. **QR code** (от Quick Response Code) — код быстрого реагирования, двухмерный или матричный штрихкод
12. **barcode** – штрихкод (из полосок разной ширины)
13. **power outlet** - точка подсоединения к сети электропитания
14. **proprietary hardware and software** – патентованное изделие и ПО
15. **seamless interaction** – бесшовное взаимодействие
16. **interoperability** - функциональная совместимость
18. **platforms out or coming out** – платформы уже существующие или готовящиеся к выходу
19. **to implement a platform** – внедрять платформу
20. **all encompassing** – всеобъемлющий
21. **to route Internet traffic** – направлять интернет поток
22. **still-prevalent [**'prev(ə)lənt] – все еще преобладающий
23. **undecillion** – ундециллион
24. **hexadecimal** – шестнадцатиричный
25. **to exceed** – превосходить
26. **interim solution** – промежуточное решение
27. **network address translation (NAT)** - трансляция сетевых адресов

Mainframe's internal structure

This article introduces one model of IBM® mainframe computer, the IBM System z9™, comparing its major hardware parts and their function to hardware parts of personal computers or notebooks.

Frames and cages

Under the covers, the mainframe has one or two metal frames, containing processor and input/output cages, as well as other physical elements.

This diagram shows the interior front view of an IBM System z9 Enterprise Class (z9 EC) model that has:

- Two frames
- One central processor complex, or CPC, cage
- Three input/output cages

24x7 design

Every hardware component inside the mainframe is designed so that the z9 can operate continuously: 24 hours a day, seven days a week.

Companies use mainframes for mission-critical work, which usually consists of applications and data that have to be available at all times. If the mainframe goes off line and access to these applications and data is suspended or lost, companies could potentially lose millions of dollars, sometimes in a matter of minutes. Mainframes also have to run efficiently, or they become too costly for companies to operate.

Although mainframes and personal computers have similar hardware parts, you will notice big differences in the quality of parts, design, and function that allow mainframes to process work continuously. Another difference is the mainframe's ability to expand its processing resources to match fluctuating or ever-increasing workloads. These built-in functions and features contribute to IBM's goal of achieving five 9s availability for its mainframes. The term 'five 9s' means that a system is available and running at least 99.999% of the time.

All IBM mainframes undergo a rigorous testing process before they are shipped to customer sites.

Central processor complex

The central processor complex, or CPC, resides in its own cage, and consists of one to four book packages. Just like its personal-computer counterpart, the motherboard or system board, each book package consists of processors, memory, timers, and input/output connections.

These collections of hardware parts are called book packages because you can slide them in or out of the CPC cage almost as easily as you can slide a book on or off a bookshelf.

Modular design

This modular packaging allows you to add or remove book packages without disrupting any work that the server is currently processing.

With System z9 models, you can add computing resources (processors, memory, or input/output connections) to your system at any time, without powering down the server.

You also can replace System z9 book packages as necessary, without powering down the server. Having this capability is like being able to change a tire on your car while it is traveling at 70 miles (or 110 kilometers) per hour!

Inside the book package

The book package consists of three distinct areas, one each for:

- The z9 EC's processors, which are inside one multichip module
- Memory cards
- Connections to input/output devices

All of the book packages plug into a backplane in the z9 EC's frame. A backplane is a circuit board that allows all connected book packages to share resources.

More about the CPC

The book package and multichip module design, as well as the design of other internal hardware, allows the z9 EC to provide continuous processing power and the capability to handle fluctuating or increasing workloads. These mainframe characteristics are known as *reliability, availability, and scalability*.

Input/output subsystem

System z9 models can have up to three input/output cages, which consist primarily of slots for input/output adapters.

Together with hardware parts in the z9 EC's central processor complex, or CPC, cage and other specialized input/output equipment on the raised floor, the hardware elements in the input/output cages constitute the high-speed subsystem that provides a path between the mainframe and peripheral devices such as printers, directors, and storage units.

The capacity of this subsystem is measured by the number of available channels, which are independent pathways that transfer data.

Channel types

Just as a personal computer has different types of bus adapters (for example, USB or SCSI), the z9 EC has different types of channels. Specific slots in the input/output cages are reserved for specific types of channels, which include the following:

•Open Systems Adapter, or OSA Express2, which provides connectivity to various industry-standard networking technologies, including Ethernet, Token Ring, and asynchronous transfer mode (known as ATM).

•Fiber Connection, or FICON® Express2, which is the most flexible channel technology. With FICON Express2, input/output devices can be located many miles or kilometers from the z9 EC to which they are attached.

•Enterprise Systems Connection, or ESCON®, which is an earlier type of fiber-optic cabling and technology. ESCON channels can provide performance almost as fast as FICON channels

Channel characteristics

Many System Z customers still use ESCON channels in their data centers. You can tell the difference between FICON (yellow) and ESCON (orange) cables in the view of cables routed from the input/output cages through the z9 EC's tailgate.

Regardless of the channel type, each z9 EC channel is an independent pathway that transfers data concurrently with other channels and the z9 EC's processors.

In contrast, the SCSI bus in a personal computer is a shared channel between all devices plugged into it. With independent channels, I/O performance on the z9 EC is significantly faster.

Design and technology

As with other hardware elements in the z9 EC, the input/output cage provides built-in redundancy; that is, the cages are designed to contain several identical functional units to prevent any interruptions in processing because of hardware failures.

Input/output capacity

Personal computers typically have a limited number of channels for input/output, but a single z9 EC can have up to 1 024 individual channels. This capacity is one factor that contributes to the mainframe's legendary scalability.

This diagram is a high-level conceptual map of input/output hardware in the mainframe environment, which shows how the z9 EC can be connected to many devices:

Input/output superhighway

The scale of mainframe input/output capacity is difficult to imagine... The number of potential devices is great enough that mainframes often use several input/output control units to offload some of the input/output processing. These control units are actually small computers that run software programs to transfer data back and forth.

To get a better idea of the mainframe's input/output capacity, just compare the connections for your personal computer to these views of typical collections of cables and switches in patch panels on the raised floor.

Close-up of FICON (yellow) and ESCON (orange) cables in a patch panel.

Internal cooling system

Mainframes rely on both internal and external cooling systems. The z9 EC has an internal cooling system that uses both air and liquid to keep its internal temperature low. The z9 EC uses liquid to cool only the processor chips; all other hardware parts, even within the book package, are cooled by air.

For the liquid cooling, the z9 EC uses the same coolant found in most of the automobiles made today. This hybrid cooling system keeps the z9 EC's processing chips at the lowest temperature in the industry. Low temperature means better performance and great reliability.

In a personal computer, a CPU heat sink draws heat off the processor; an internal fan forces cool air across the heat sink, and forces hot air out of the computer case. Similarly, the z9 EC has a heat sink and a sophisticated arrangement of fans to circulate air through the frame. Sensors in the cooling system monitor and control fan speeds to maintain the internal temperature within its optimum range.

External cooling system

A typical data center has a raised floor, that is devoted to housing mainframes and other computing equipment. A bank of mainframes blowing air out into the raised floor can make even a large room hot, so raised floor areas require air-conditioning to dissipate the excess heat.

The raised floor and the crawl space below it are both cooled with powerful air-conditioning, and air from the crawl space keeps the mainframe cool internally as well.

When a mainframe is positioned on the raised floor, the solid floor tiles on that spot are replaced with perforated tiles; some of the openings in the tiles are for cabling to be routed from the main frame into the crawl space, but other perforations allow cool air from the crawlspace to blow up inside the frame.

Powering the mainframe

Personal computers and notebooks use power supplies and batteries for their operation; the z9 EC also has power supplies and batteries that provide similar functions.

A key difference in the z9 EC is the redundancy that is required for a state-of-the-art mainframe system to provide continuous availability. Redundancy is the use of several identical functional units to prevent any interruptions in processing because of hardware failures.

In the z9 EC, two internal bulk power units provide power from an external source; the electrical current is delivered through two independent cables attached to the power units.

These power units are designed so that the z9 EC can continue to operate even if one of the units fails.

Power supply and battery design

The power units also are designed to work with the internal batteries in such a way that, if the power supply from the external source is interrupted, the z9 EC can switch from the power units to internal batteries without any loss of work.

The z9 EC also has subsidiary power units in the central processor complex cage and the input/output cages. These subsidiary units provide independent power so that, for example, one book package can be removed from the central processor complex cage without shutting down power to the other book packages.

Support elements

The z9 EC has two support elements that constitute part of the hardware control system for the mainframe. These support elements are Lenovo™ ThinkPads™ that are used primarily by IBM customer engineers (known as CEs), who install, start and service mainframes on the raised floor.

Customers use a standard Web browser to access the Hardware Master Console, or HMC, which is used to operate and monitor the z9 EC. Like the support elements, HMCs also may be ThinkPads or another type of computer.

Hardware control system

The two support elements on the z9 EC constitute only part of the mainframe's hardware control system. The other elements of this control system include:

•Flexible support processors, which are IBM microprocessors located inside each power supply, each book package, and each input/output cage in the z9 EC's frames.

• Two internal Ethernet® local area networks, or LANs, which connect the support elements to all of the flexible support processors. The z9 EC has two internal LANs and two support elements for redundancy; if one LAN or support element fails, the hardware control system can be dynamically switched over to the backup LAN or support element.

•An external Ethernet LAN, which connects both of the z9 EC's support elements to one or more Hardware Master Consoles, or HMCs, through which customers operate and monitor the z9 EC.

Active vocabulary

1. **built-in redundancy -** запланированная избыточность
2. **to off-load input-output processing** – передать другой вычислительной системе часть функций по обработке данных ввода-вывода
3. **patch panel** - коммутационная (распределительная) панель, содержащая набор гнёзд для соединения устройств в шкафу с сетевым оборудованием
4. **close-up –** фото крупным планом
5. **coolant** – охлаждающая жидкость
6. **CPU heat sink** – радиатор процессора
7. **to draw the heat off** – отводить тепло
8. **to dissipate the excess heat** – рассеивать избыточное тепло
9. **raised floor and the crawl space** – фальшпол и полупроходное техническое подполье (высотой до 1,5 м)
10. **solid and perforated tiles** – сплошные и перфорированные пластины пола
11. **power supply** – электроснабжение
12. **state-of-the-art mainframe** – высококачественная вычислительная машина
13. **internal bulk power unit** – внутренний блок магистрального электроснабжения
14. **to switch from the power units to internal batteries** – переключиться от блоков питания к внутренним аккумуляторам
15. **subsidiary power unit** – дополнительный блок питания
16. **support element** – устройство для технического обслуживания
18. **customer engineer (CE)** - инженер по эксплуатации, ремонту и профилактическому обслуживанию
19. **flexible processor** – гибко-программируемый процессор
20. **LAN (Local area network) Ethernet** – локальная сеть передачи данных по протоколу канального уровня / в среде Ethernet

1. **mainframe -** большая многопользовательская вычислительная система, суперкомпьютер
2. **frame** – (металическая) рама, шасси
3. **input/output cage** – коробка для устройств ввода/вывода
4. **mission-critical systems** - системы непрерывного действия, системы с непрерывным доступом к данным, системы для решения критически важных/боевых, ответственных задач
5. **to suspend access** – временно прекратить доступ

6. **fluctuating workload** – меняющаяся рабочая нагрузка
7. **to undergo a rigorous testing** – подвергаться жесткому контролю
8. **to reside in a cage** – располагаться в коробке
9. **personal-computer counterpart** – соответствующий по функции элемент ПК
10. **book** – логический блок, том, процессорный пакет (термин IBM)
11. **to disrupt the work** – нарушить работу
12. **computing resources** – вычислительные ресурсы
13. **to power down the server** – обесточить сервер, отключить питание
14. **EC (Enterprise class)** – для крупных корпораций
15. **backplane** – коммутационная панель, кросс-шина
16. **scalability** – масштабируемость
17. **raised floor** – фальшпол, технологический пол, устанавливаемый в больших вычислительных центрах, под которым проходят все кабели. Из-за высокой стоимости в настоящее время применяется крайне редко
18. **director** – блок управления
19. **bus adapter** – шинный адаптер
20. **SCSI [**'skʌzɪ**] (small computer system interface)** - интерфейс малых компьютерных систем
21. **Token Ring** – сеть передачи данных с маркерным доступом (использующая передачу маркера как символа права доступа к сети) (термин IBM)
23. **Fiber Connection** – оптоволоконное соединение
24. **tailgate** – нижний люк
25. **concurrently** - одновременно, параллельно

How caching works

The CPU accesses memory according to a distinct hierarchy. Whether it comes from permanent storage (the hard drive) or input (the keyboard), most data goes in **random access memory** (RAM) first. The CPU then stores pieces of data it will need to access, often in a **cache**, and maintains certain special instructions in the **register**.

There are memory caches, hardware and software disk caches, page caches and more. Virtual memory is even a form of caching.

Caching is a technology based on the memory subsystem of a computer. The main purpose of a cache is to accelerate computer while keeping the price of the computer low.

Caching implies the use of a faster but smaller memory type to accelerate a slower but larger memory type.

When using a cache, you must check the cache first to see if an item is in there. If it is there, it's called a **cache hit**. If not, it is called a **cache miss** and the computer must wait for a round trip from the larger, slower memory area.

Computer caches

A computer is a machine in which we measure time in very small increments. When the microprocessor accesses the main memory, it does it in about 60 nanoseconds (60 billionths of a second). That's pretty fast, but it is much slower than the typical microprocessor. Microprocessors can have cycle times as short as 2 nanoseconds, so to a microprocessor 60 nanoseconds seems like an eternity.

What if we build a special memory bank in the motherboard, small but very fast (around 30 nanoseconds)? That's already two times faster than the main memory access. That's called a level 2 cache or an **L2 cache**. What if we build an even smaller but faster memory system directly into the microprocessor's chip? That way, this memory will be accessed at the speed of the microprocessor and not the speed of the memory bus. That's an **L1 cache**, which on a 233-megahertz (MHz) Pentium is 3.5 times faster than the L2 cache, which is two times faster than the access to main memory.

Some microprocessors have two levels of cache built right into the chip. In this case, the motherboard cache -- the cache that exists between the microprocessor and main system memory -- becomes level 3, or **L3 cache**.

There are a lot of subsystems in a computer; you can put cache between many of them to improve performance. Here's an example. We have the microprocessor (the fastest thing in the computer). Then there's the L1 cache that caches the L2 cache that caches the main memory which can be used (and is often used) as a cache for even slower peripherals like hard disks and CD-ROMs. The hard disks are also used to cache an even slower medium -- your Internet connection.

Your **Internet connection** is the slowest link in your computer. So your browser (Internet Explorer, Opera, etc.) uses the hard disk to store HTML pages, putting them into a special folder on your disk. The first time you ask for an HTML page, your browser renders it and a copy of it is also stored on your disk. The next time you request access to this page, your browser checks if the date of the file on the Internet is newer than the one cached. If the date is the same, your browser uses the one on your hard disk instead of downloading it from Internet. In this case, the smaller but faster memory system is your hard disk and the larger and slower one is the Internet.

To give you the big picture of it all, here's a list of a normal caching system:

- **L1 cache** - Memory accesses at full microprocessor speed (10 nanoseconds, 4 kilobytes to 16 kilobytes in size)
- **L2 cache** - Memory access of type SRAM (around 20 to 30 nanoseconds, 128 kilobytes to 512 kilobytes in size)

 Main memory - Memory access of type RAM (around 60 nanoseconds, 32 megabytes to 128 megabytes in size)
 Hard disk - Mechanical, slow (around 12 milliseconds, 1 gigabyte to 10 gigabytes in size)
 Internet - Incredibly slow (between 1 second and 3 days, unlimited size)

As you can see, the L1 cache caches the L2 cache, which caches the main memory, which can be used to cache the disk subsystems, and so on.
It is kind of incredible that relatively tiny amounts of memory can maximize the use of much larger amounts of memory. Think about a 256-kilobyte L2 cache that caches 64 megabytes of RAM. In this case, 256,000 bytes efficiently caches 64,000,000 bytes. Why does that work?
In computer science, we have a theoretical concept called **locality of reference**. It means that in a fairly large program, only small portions are ever used at any one time. As strange as it may seem, locality of reference works for the huge majority of programs. Even if the executable is 10 megabytes in size, only a handful of bytes (actually about 5%) from that program are in use at any one time, and their rate of repetition is very high.
Source: https://computer.howstuffworks.com/cache.htm
1. **to access memory** – обращаться к памяти
2. **permanent storage** – постоянное место хранения
3. **cache** [kæʃ] – кэш, сверхоперативная память
4. **register** – регистр, устройство сверхбыстродействующей памяти в процессоре, служащее для временного хранения команд, управляющей информации, операндов и/или результатов выполняемых операций.
5. **hardware and software caches** – аппаратные и программные кэши
6. **page cache** - страничный кэш, в котором в виртуальной памяти
запоминаются страницы, а не блоки данных, как в дисковом кэше
7. **caching** – кэширование, работа с использованием кэша, подкачка в кэш
8. **cache hit** – удачное обращение к кэш-памяти, "попадание", наличие нужных данных (в кэше)
9. **cache miss** – неудачное обращение к кэш-памяти, "промах, непопадание", отсутствие нужных данных [в кэше]
10. **increment** – (бесконечно малое) приращение
11. **level 2 cache** – кэш второго уровня
12. **memory bus** – шина памяти шина, соединяющая ЦПУ и главную память
13. **slower medium** – более медленная среда передачи данных
14. **HTML page (hypertext markup language**) - страница гипертекстового документа, веб-страница
15. **folder** – папка (на диске)
16. **to render a page** – отображать страницу
17. **to request access to** – запросить доступ к (странице)
18. **SRAM (static random access memory)** - статическая оперативная память
19. **locality of reference** – локальность ссылок, предсказуемость программных запросов к данным во времени и пространстве
20. **executable** *n* – исполняемый файл, *exe*-файл
21. **rate of repetition** – частота повторений

How BIOS works

One of the most common uses of Flash memory is for the basic input/output system of your computer, commonly known as the BIOS. The BIOS software has a number of different roles, but its most important role is to load the operating system. When you turn on your computer and the microprocessor tries to execute its first instruction, it has to get that instruction from somewhere. It cannot get it from the operating system because the operating system is located on a hard disk, and the microprocessor cannot get to it without some instructions that tell it how. The BIOS provides those instructions. Some of the other common tasks that the BIOS performs include:

A power-on self-test (POST) for all of the different hardware components in the system to make sure everything is working properly

Activating other BIOS chips on different cards installed in the computer - For example, SCSI and graphics cards often have their own BIOS chips.

Providing a set of low-level routines that the operating system uses to interface to different hardware devices - It is these routines that give the BIOS its name. They manage things like the keyboard, the screen, and the serial and parallel ports, especially when the computer is booting.

Managing a collection of settings for the hard disks, clock, etc.

The BIOS is special software that interfaces the major hardware components of your computer with the operating system. It is usually stored on a Flash memory chip on the motherboard, but sometimes the chip is another type of ROM.

When you turn on your computer, the BIOS does several things. This is its usual sequence:

1. Check the CMOS Setup for custom settings
2. Load the interrupt handlers and device drivers
3. Initialize registers and power management
4. Perform the power-on self-test (POST)
5. Display system settings
6. Determine which devices are bootable
7. Initiate the bootstrap sequence

The first thing the BIOS does is check the information stored in a tiny (64 bytes) amount of RAM located on a complementary metal oxide semiconductor (CMOS) chip. The CMOS Setup provides detailed information particular to your system and can be altered as your system changes. The BIOS uses this information to modify or supplement its default programming as needed.

Interrupt handlers are small pieces of software that act as translators between the hardware components and the operating system. For example, when you press a key on your keyboard, the signal is sent to the keyboard interrupt handler, which tells the CPU what it is and passes it on to the operating system. The device drivers are other pieces of software that identify the base hardware components such as keyboard, mouse, hard drive and floppy drive. Since the BIOS is constantly intercepting signals to and from the hardware, it is usually copied, or shadowed, into RAM to run faster.

Occasionally, a computer will need to have its BIOS updated. This is especially true of older machines. As new devices and standards arise, the BIOS needs to change in order to understand the new hardware. Since the BIOS is stored in some form of ROM, changing it is a bit harder than upgrading most other types of software. 18

Source: https://computer.howstuffworks.com/bios.htm

1. **Flash memory** – память, сохраняющая свое содержимое без питания и регенерации
2. **BIOS (basic input/output system)** - базовая система ввода/вывода, БИОС
3. **commonly** – обычно
4. **provide v** – предоставлять

5. **POST (Power-On Self Test)** - самотестирование при включении питания и рестарте нажатием на кнопку. При теплом (программном) рестарте осуществляется сокращенное самотестирование
6. **make sure v** - убедиться
7. **SCSI (Small Computer Systems Interface)** – интерфейс малых вычислительных систем, интерфейс "скази", тип параллельного интерфейса "быстрых" периферийных устройств
8. **low-level routines** – низко-уровневые программы, подпрограммы
9. **interface (to, with) v** – подключать, соединять с
10. **manage v** – управлять
11. **boot (up) v** – загружать(ся)
12. **custom settings** – пользовательские установки
13. **be stored v** – храниться
14. **ROM (Read Only Memory)** – постоянное запоминающее устройство
15. **turn on** – включать
16. **sequence** – последовательность
17. **CMOS (complementary metal oxide semiconductor)** - комплементарный металло-оксидный полупроводник
18. **interrupt handler** - обработчик прерываний, подпрограмма обработки прерываний
19. **device driver** - драйвер устройства
20. **register** – регистр, устройство сверхбыстродействующей памяти в процессоре, служащее для временного хранения управляющей информации, операндов и/или результатов выполняемых данной микросхемой операций.
21. **power management (PM)** - управление [электро]питанием
22. **initialize v -** инициализировать, задать начальное значение переменной
23. **determine v** – определять
24. **initiate v** - инициировать; запускать; включать
25. **bootstrap sequence** - программа самозагрузки, программа раскрутки
26. **tiny** – маленький, крошечный
27. **Setup** - программа или процедура начальной установки
28. **alter v** – изменять
29. **particular** – особый, специфичный
30. **supplement v** – дополнить
31. **intercept v** - перехватывать
32. **shadow v** – точно копировать

How Motes Work

Technology of **motes,** also called **smart dust** and **wireless sensing networks,** is a paradigm for distributed sensing and it is opening up a fascinating new way to look at computers.
The basic idea of the “mote” concept is pretty simple:
 The core of a mote is a small, low-cost, low-power **computer**.
 The computer monitors one or more **sensors**. It is easy to imagine all sorts of sensors, including sensors for temperature, light, sound, position, acceleration, vibration, stress, weight, pressure, humidity, GPS etc.
 The computer connects to the outside world with a **radio link**. The most common radio links allow a mote to transmit at a distance of something like 5-100 meters. Power consumption, size and cost are the barriers to longer distances. Since a fundamental concept with motes is tiny size (and associated tiny cost), small and low-power radios are normal.

Motes can either run off of batteries, or they can tap into the power grid in certain applications. As motes shrink in size and power consumption, it is possible to imagine solar power or even something exotic like **vibration power** to keep them running.
In March, 2003, researchers managed to cram all of the parts needed for a mote onto a single chip less than 3 millimeters on each side. The total size is about 5 square millimeters, meaning that you could fit more than a dozen of these chips onto a penny. A single-chip "Spec" mote has an AVR-like RISC core, 3K of memory, an 8-bit, on-chip ADC, an FSK radio transmitter, a Paged memory system, communication protocol accelerators, register windows, and much, much more.

Typical Applications

You could embed motes in bridges when you pour the concrete. The mote could have a sensor on it that can detect the salt concentration within the concrete. They may be recharged periodically by electromagnetic radiation. Other possible sensors might detect vibration, stress, temperature swings, cracking, etc., all of which would help maintenance personnel spot problems long before they become critical.
You could connect sensors to a mote that can monitor the condition of machinery -- temperature, number of revolutions, oil level, etc. and log it in the mote's memory. The mote could transmit all the logged data. This would allow detailed maintenance records to be kept on machinery (for example, in an oil field), without maintenance personnel having to go measure all of those parameters themselves.
However, much of the greatest excitement about motes comes from the idea of using large numbers of them forming **ad hoc networks,** where individual motes communicate with each other and pass data along from one another and further to a bigger processing center.
Imagine a suburban neighborhood or an apartment complex with motes that monitor the water and power meters. Since all of the meters (and motes) in a typical neighborhood are within 30 meters of each other, the attached motes could form an ad hoc network amongst themselves. At one end of the neighborhood is a super-mote with a network connection or a cell-phone link. Measurement can occur hourly or daily if desired.
A farmer, vineyard owner, or ecologist could equip motes with sensors that detect temperature, humidity, etc., making each mote a mini weather station. Scattered throughout a field, vineyard or forest, these motes would allow the tracking of micro-climates.
A building manager could attach motes to every electrical wire throughout an office building. These motes would have induction sensors to detect power consumption on that individual wire and let the building manager see power consumption down to the individual outlet. If power consumption in the building seems high, the building manager can track it to an individual tenant.

A biologist could equip an endangered animal with a collar containing a mote that senses position, temperature, etc. As the animal moves around, the mote collects and stores data from the sensors. In the animal's environment, the biologists could place zones or strips with data collection motes. When the animal wanders into one of these zones, the mote in the collar would dump its data to the ad hoc network in the zone, which would then transmit it to the biologist.

1. **mote (sensor)** – сверхминиатюрный датчик, "интеллектуальная пылинка", датчик с автономным питанием (может работать и на солнечной энергии)
2. **smart dust** – интеллектуальная пыль, системы интеллектуальных датчиков для сбора данных в трудно доступных местах
3. **wireless sensing networks** - беспроводная сенсорная сеть
4. **paradigm** – концепция
5. **distributed sensing** – распределенная сенсорная сеть
6. **acceleration** – ускорение
7. **stress** – (механическое) напряжение
8. **humidity** – влажность
9. **button battery** - батарея таблеточного типа
10. **onboard flash memory** – встроенная флэш память
12. **to run of power grid** – работать от сети
13. **to shrink in size** – уменьшаться в размере
14. **solar power** – солнечная энергия
15. **to cram into** – втиснуть
17. **RISC (Reduced Instruction Set Computing) (technology)** – технология вычисления с сокращённым набором команд, (их почти в три раза меньше, чем у аналогичных по разрядности обычных процессоров)
18. **ADC (analog-to-digital converter)** - аналогово-цифровой преобразователь
19. **FSK (frequency shift keying) radio transmitter** – радио передатчик с частотной манипуляция
20. **paged memory** – страничная память
21. **accelerator** – ускоритель
22. **register windows** – регистровые окна служащие для повышения эффективности вызова процедур и переключения с задачи на задачу
23. **to embed motes** – включать сенсоры в структуру материала
24. **to pour concrete** – заливать бетон
25. **temperature swings** – колебания температуры
26. **cracking** – образование трещин
27. **number of revolutions** – количество вращений
28. **to log in memory** – записывать в памяти
29. **oil field** – нефтяное месторождение
30. **maintenance personnel** – персонал, осуществляющий техобслуживание
31. **ad hoc network** – специальная сеть; беспроводная локальная сеть, состоящая из станций без точки доступа
32. **water and power meters** – счетчики потребления воды и электричества
33. **induction sensor** – индукционный датчик
34. **power consumption** – потребление электричества
35. **individual outlet** – отдельная штепсельная розетка
36. **endangered animal** – исчезающее животное
37. **to dump data** – выгружать данные

How Computer Keyboards Work

Keyboards use a variety of switch technologies. It is interesting to note that we generally like to have some audible and **tactile** response to our typing on a keyboard. We want to hear the keys "click" as we type, and we want the keys to feel firm and spring back quickly as we press them. Let's take a look at these different technologies:

Probably the most popular switch technology in use today is **rubber dome**. In these keyboards, each key sits over a small, flexible rubber dome with a hard carbon center. When the key is pressed, a plunger on the bottom of the key pushes down against the dome. This causes the carbon center to push down also, until it presses against a hard flat surface beneath the key matrix. As long as the key is held, the carbon center completes the circuit for that portion of the matrix. When the key is released, the rubber dome springs back to its original shape, forcing the key back up to its at-rest position. Rubber dome switch keyboards are inexpensive, have pretty good tactile response and are fairly resistant to spills and corrosion because of the rubber layer covering the key matrix. **Membrane** switches are very similar in operation to rubber dome keyboards. A membrane keyboard does not have separate keys though. Instead, it has a single rubber sheet with bulges for each key. You have seen membrane switches on many devices designed for heavy industrial use or extreme conditions. Because they offer almost no tactile response and can be somewhat difficult to manipulate, these keyboards are seldom found on normal computer systems.

Capacitive switches are considered to be non-mechanical because they do not simply complete a circuit like the other keyboard technologies. Instead, current is constantly flowing through all parts of the key matrix. Each key is spring-loaded, and has a tiny plate attached to the bottom of the plunger. When a key is pressed, this plate is brought very close to another plate just below it. As the two plates are brought closer together, it affects the amount of current flowing through the matrix at that point. The processor detects the change and interprets it as a keypress for that location. Capacitive switch keyboards are expensive, but do not suffer from corrosion and have a longer life than any other keyboard. Also, they do not have problems with bounce since the two surfaces never come into actual contact.

Metal contact and **foam element** keyboards are not as common as they used to be. Metal contact switches simply have a spring-loaded key with a strip of metal on the bottom of the plunger. When the key is pressed, the metal strip connects the two parts of the circuit. The foam element switch is basically the same design but with a small piece of spongy foam between the bottom of the plunger and the metal strip, providing for a better tactile response. Both technologies have good tactile response, make satisfyingly audible "clicks" and are inexpensive to produce. The problem is that the contacts tend to wear out or corrode faster than on keyboards that use other technologies. Also, there is no barrier that prevents dust or liquids from coming in direct contact with the circuitry of the key matrix.

Source: https://computer.howstuffworks.com/keyboard.htm

1. **audible** – слышимый
2. **tactile response** – осязаемый отклик
3. **click v** – щелкать
4. **key** – клавиша
5. **keyboard** – клавиатура
6. **spring back v** – отскакивать назад, возвращаться в исходное положение
7. **rubber dome switch** – переключатель с резиновым куполом
8. **flexible** – гибкий
9. **carbon** – угольный
10. **plunger on the bottom** – шток на основании (клавиши)
11. **key matrix** – переключающая матрица

12. **complete the circuit v** – замыкать цепь
13. **release v** – отпускать (клавишу)
14. **at rest position** – положение в состоянии покоя
15. **fairly resistant to** …- довольно устойчивый к…
16. **spill** – пролитая жидкость
17. **membrane switch** – мембранный переключатель
18. **rubber sheet** – лист резины, клеенка
19. **bulge** – округлый выступ, выпуклость
20. **extreme conditions** – неблагоприятные условия
21. **capacitive switch** – емкостной ключ, переключатель
22. **current** – электрический ток
23. **spring-loaded** – снабженный пружиной
24. **attach v** – прикреплять
25. **plate** – пластина
26. **keypress** – нажатие клавиши
27. **bounce** – дрожание (матрицы от удара клавиш), приводящее к колебанию напряжения
28. **metal contact switches** – переключатели с металлическими контактами
29. **strip of metal** – полоска металла
30. **spongy foam** – губчатая (полимерная) пена
31. **provide for** – обеспечивать
32. **tend to v** – иметь склонность к
33. **wear out v** - изнашиваться
34. **circuitry** – схема (электрическая)

What are relational databases?

A **database** is an organized collection of data, stored and accessed electronically. Database designers typically organize the data to model aspects of reality in a way that supports processes requiring information, such as (for example) modeling the availability of rooms in hotels in a way that supports finding a hotel with vacancies. Formally, a "database" refers to a set of related data and the way it is organized. Access to this data is usually provided by a "database management system" (DBMS) consisting of an integrated set of computer software that allows users to interact with one or more databases and provides access to all of the data contained in the database (although restrictions may exist that limit access to particular data). The DBMS provides various functions that allow entry, storage and retrieval of large quantities of information and provides ways to manage how that information is organized.

Databases have been a staple of business computing from the very beginning of the digital era. In fact, the **relational database** was born in 1970 when E.F. Codd, a researcher at IBM, wrote a paper outlining the process. Since then, relational databases have grown in popularity to become the standard.

Originally, databases were flat. This means that the information was stored in one long text file, called a **tab delimited file**. Each entry in the tab delimited file is separated by a special character, such as a vertical bar (**|**). Each entry contains multiple pieces of information (fields) about a particular object or person grouped together as a record. The text file makes it difficult to search for specific information or to create reports that include only certain fields from each record.

You have to search sequentially through the entire file to gather related information, such as age or salary. A relational database allows you to easily find specific information. It also allows you to sort based on any field and generate reports that contain only certain fields from each record.

Relational databases use **tables** to store information. The standard fields and records are represented as columns (fields) and rows (records) in a table.

With a relational database, you can quickly compare information because of the arrangement of data in columns. The relational database model takes advantage of this uniformity to build completely new tables out of required information from existing tables. In other words, it uses the relationship of similar data to increase the speed and versatility of the database.

The "relational" part of the name comes into play because of mathmatical relations. A typical relational database has anywhere from 10 to more than 1,000 tables. Each table contains a column or columns that other tables can key on to gather information from that table.

By storing this information in another table, the database can create a single small table with the locations that can then be used for a variety of purposes by other tables in the database. A typical large database, like the one a big Web site, such as Amazon would have, will contain hundreds or thousands of tables like this all used together to quickly find the exact information needed at any given time.

Relational databases are created using a special computer language, **structured query language (SQL)**, that is the standard for database interoperability. SQL is the foundation for all of the popular database applications available today, from **Access** to **Oracle**.

1. **Staple** – скоба
2. **tab delimited file** - файл с разделителями табуляцией
3. **flat** – одномерный
4. **vertical bar** - вертикальная черта
5. **field** – поле
6. **record** – запись
7. **sequentially** - последовательно
8. **relational database** – реляционная база данных

9. **arrangement** – расположение
10. **uniformity** – единообразие
11. **versatility** — переменчивость, гибкость
12. **SQL** - язык структурированных запросов
13. **interoperability** - совместимость

How do web conferencing programs work?

Real-time text chat facilities such as IRC appeared in the late 1980s. Web-based chat and instant messaging software appeared in the mid-1990s. The PLATO computer learning system allowed students to collaborate on networked computers to accomplish learning tasks as early as the 1960s, but the early networking was not accomplished via the World Wide Web and PLATO's collaborative goals were not consistent with the presenter-audience dynamic typical of web conferencing systems.

Several point-to-point and private-network video conferencing products were introduced in the early 1990s,most notably CU-SeeMe, which was used to link selected schools around the United States of America in real-time collaborative communications as part of the Global Schoolhouse project from Global SchoolNet.

In May 1995, PictureTel announced LiveShare Plus as a general-use data collaboration product for Windows-based personal computers. The software allowed application sharing, user-granted control of a remote PC, shared whiteboard markup, file transfer, and text messaging. List price was given as $249 per computer. PictureTel referenced an agreement with Microsoft in its announcement press release, and a May 26, 1995 memo from Bill Gates to Microsoft executive staff and direct reports said "Our PictureTel screen sharing client allowing Window sharing should work easily across the Internet." A wide variety of web conferencing programs are available on the market. The simplest use chat and instant messaging programs host text-based group discussions. More sophisticated programs exchange visual information using webcams and streaming video and allow people to share documents online. Some programs are entirely computer- and Internet-based, and others use the telephone system to distribute audio content.

Web conferencing programs combine tools already common to web pages and Internet communication. They bundle these tools into one interface to create an interactive meeting environment. These tools include:

- HTML, XML and ASP markup
- Java scripts
- Flash animation
- instant messaging
- Streaming audio and video

Companies can either **purchase** conferencing software and host their meetings themselves or use a **hosting service**. Hosting services provide the software and server space on which to conduct meetings. Either way, the company or the hosting service must have software to coordinate the meeting and ample server space and bandwidth to accommodate it.

Some programs can merge with a company's existing e-mail, calendar, messaging and office productivity applications. Some allow attendees to view the presentation in their regular web browser without installing any additional software. Depending on the software, people can:

- View slide presentations from programs like PowerPoint.
- Draw or write on a common **whiteboard** by using their computer mice or typing.
- Transmit still pictures or video to other attendees via a **webcam**. (This increases the required bandwidth and can sometimes slow the transfer of the presentation.)
- View information from the moderator's computer desktop using screen sharing.
- Share documents, often even if attendees don't have the software that created them, using **application sharing**.
- Ask and answer questions through audio chat (as an integrated part of the software) or by phone.

Programs include options for security and encryption. Most require moderators and attendees to use a login name and password to access the meeting. Some use SSL or TLS encryption to

protect data. Some companies host web conferences on internal servers so that the data stays behind the corporate firewall.
Source: https://money.howstuffworks.com/business-communications/question772.htm

1. **Host** – быть местом проведения
2. **instant messaging** – программы мгновенного обмена сообщениями
3. **streaming video** – потоковое воспроизведение
4. **entirely** — полностью
5. **distribute** — распространять
6. **bundle** – связывать, объединять
8. **markup** – разметка
9. **purchase** - приобретать
10. **ample** — обильный
11. **bandwidth** — полоса пропускание
12. **Merge** – сливаться, объединяться
13. **attendee** — участник, посетитель
14. **screen sharing** – разделенный экран
15. **integrated** – встроенный
16. **encryption** — шифрование
17. **internal** - внутренний

HTML5

HTML5 is a markup language used for structuring and presenting content on the Internet. It is the fifth and current major version of the HTML standard. It currently exists in two standardized forms: *HTML 5.2 Recommendation* by the World Wide Web Consortium (W3C), intended primarily for Web content developers; and *HTML Living Standard* by WHATWG, intended primarily for browser developers. The HTML that we use today has been developed from a prototype written by Tim Berners-Lee in 1992.

In 1989, Tim Berners-Lee invented the Web with HTML as its publishing language. It was at CERN, the European Organization for Nuclear Research, where the World Wide Web was born. Initially, HTML was based on Standard Generalized Markup Language (SGML), a standard for marking up text into structural units like paragraphs, lists, etc. in a document.

However, with the evolution of the World Wide Web (WWW), HTML began to proliferate and rapidly dispersed its roots into the mainstream. Soon, the browser war began and various companies started to create their own browsers. As a result, HTML became fragmented and some web pages appeared fine in one browser but not in another.

In 1994, the Internet Engineering Task Force (IETF) created an HTML Working Group. This resulted in the birth of HTML 2.0 in September 1995 followed by HTML 3.2 in January 1997, HTML 4.01 in December 1999, HTML5 in October 2014 and HTML5.1 in November 2016.

Released in October 2014, HTML5 was developed with the intent to improve the language with support for the latest interactive multimedia, while keeping it both readable by humans and seamlessly understandable by computers and web devices. The new language incorporates features like video playback, drag-and-drop, etc., which were not present on the previous version and we had to depend on third-party browser plug-ins such as Adobe Flash, Google Gears, and Microsoft Silverlight.

HTML5 incorporates several new features in comparison to the previous versions of HTML. It includes detailed processing models to energize more interoperable usage; it extends, enhances and rationalizes the markup available for archives, and provides markup and application programming interfaces (APIs) for complex web applications. HTML5 is also a contender for cross-platform mobile applications as the newly incorporated features in the latest version of HTML are designed to cater to the low-powered devices.

HTML and HTML5 have a lot in common. However, HTML5 has a few new features incorporated in it:

New Semantic Elements − HTML5 includes a few content-specific elements like <header>, <footer>, <article>, <nav>, and <section>. <footer> is used for a footer of a document or section, could include the name of the author, the date of the document, contact information, or copyright information. <header> is for an introduction of a document or section. <main> is for main content area consists of content that is directly related to, or expands upon the central topic of a document or the central functionality of an application. <mark> is used for text that should be highlighted. <menu> is for list of commands used in toolbar, popup and other menu. <section> is for a section in a document (chapters, headers, footers, or any other sections of the document). <article> is used for external content, like text from a news-article, blog, forum, or any other content from an external source. <aside> is for content aside from the content it is placed in. The aside content should be related to the surrounding content. **Persistent Local Storage** − HTML uses cookies or browser cache as temporary storage while HTML5 provides local storage in the form of **Application Cache**, **Web SQL database**, and **Web storage**.

WebSocket − This is a next-generation bidirectional communication technology meant for web applications.

Server-Sent Events − HTML5 brings in events, called Server-Sent Events (SSE), which flow from the web server to the web browsers.

Canvas − A new element that supports two-dimensional drawing and can be programmed with JavaScript.
Audio and Video − HTML4 does not include audio and video specification. Therefore, it necessitates the use of third-party plugins to embed audio and video on web pages. Being an enhanced version of HTML4, HTML5 includes audio and video specifications as an integral part of its features. HTML5 allows embedding audio or video on web pages without the need of using any third-party plugins. We use <audio> to represent a sound, or an audio stream; <embed> for embedded objects in the document; <video> for a video, and its associated audio files and captions, with the necessary interface to play it.
Geolocation − In HTML4, it is difficult to detect the accurate geolocation of users when browsing web pages (especially on mobile devices). When it comes to HTML5, the JS GeoLocation API helps identify the true location of users browsing any website. HTML5 supports geolocation tagging enabling visitors to share their physical location via their web applications.
Drag and Drop − HTML5 lets you drag and drop items from one location to another location on the same web page. **Vector Graphics** – HTML requires third-party technologies like VML, Silverlight, Flash, etc. for vector graphics while HTML5 allows embedding vector graphics, e.g. SVG. <canvas> represents a bitmap area that scripts can be used to render graphics, like graphs, game graphics, or any visual images on the fly. Tag <svg> defines an embedded vector image.
An opportunity to add a custom-made element into an HTML5 document is also presented in HTML5. The JavaScript method document.registerElement() may used to achieve that. This method must be passed as the name of the custom element that will be used as well as with an optional object that defines the API. The following example shows how to create a custom element <custom-elm> and add it to existing page:
var myCustomElm = document.registerElement(custom-elm);
document.body.appendChild(new myCustomElm());
The end result would be a new element inside our document's body (<body>):
<custom-elm></custom-elm>
Source: https://www.springpeople.com/blog/web-development-basics-what-is-the-difference-between-html-and-html5/

1. **markup language** – язык разметки
2. **to proliferate** – быстро распространяться
3. **to disperse** – распределять, рассредоточивать
4. **playback** – проигрывание
5. **drag-and-drop** – с поддержкой перетаскивания объектов мышью
6. **third-party browser plug-ins** – сторонние дополнительные модули для браузера
7. **to incorporate** – включать в себя
8. **interoperable** – взаимодействующий
9. **to enhance** – расширять, совершенствовать
10. **application programming interfaces** – программный интерфейс приложения (набор классов и прототипов членов-функций; интерфейс содержит такие данные о функции, как ее имя, типы параметров, типы возвращаемых значений и генерируемые исключения)
11. **to cater** – снабжать, ориентироваться
12. **low-powered device** – устройства малой мощности
13. **content-specific elements** – определяемые содержанием элементы
14. **to highlight** – выделять
15. **local storage** – локальное запоминающее устройство
16. **footer** – низ страницы
17. **header** – верх страницы

18. **chapter** – раздел, часть
19. **external source** – внешний источник
20. **application cache** – кэш приложения
22. **Server-Sent Events** – события, посылаемые сервером
23. **to flow** – поступать от чего-либо
24. **canvas** – холст
25. **two-dimensional** – двумерный
26. **embedded objects** – встроенный объект
27. **geolocation tagging** – геолокация
28. **bitmap** – растровые изображения
29. **to render graphics** – выполнять отрисовку изображения
30. **custom-made element** – изготовленный по специальным техническим требованиям заказчика

Basics of Wired Networks

Networking allows one computer to send information to and receive information from another. The Internet is the most conspicuous example of computer networking, linking millions of computers around the world, but smaller networks play a role in information access on a daily basis. Networking allows many different devices in multiple locations to access a shared repository of data. Networks also allow users in one locality to share printers, other peripherals and disk-systems among the networked units.

Networks can be categorized in several different ways. One approach defines the type of network according to the geographic area it spans. Here we can classify network technologies as belonging to one of two basic groups. Local area network (LAN) technologies connect many devices that are relatively close to each other, usually in the same building. The library terminals that display book information would connect over a local area network. Wide area network (WAN) technologies connect a smaller number of devices that can be many kilometers apart. For example, if two libraries at the opposite ends of a city wanted to share their book catalog information, they would most likely make use of a wide area network technology, which could be a dedicated line leased from the local ISP, intended solely to carry their data.

LAN and WAN were the original categories of area networks, while the others have gradually emerged over many years of technology evolution. Here are the most common examples:

WLAN - Wireless Local Area Network

WAN - Wide Area Network

MAN - Metropolitan Area Network

SAN - Server Area Network, or sometimes Small Area Network

CAN - Campus Area Network

PAN - Personal Area Network

DAN - Desk Area Network

The Ethernet is a local area technology, connecting devices in close proximity. In 1973, at Xerox Corporation’s Palo Alto Research Center (more commonly known as PARC), researcher Bob Metcalfe designed and tested the first Ethernet network. While working on a way to link Xerox’s "Alto" computer to a printer, Metcalfe developed the physical method of cabling that connected devices on the Ethernet as well as the standards that governed communication on the cable. Ethernet has since become the most popular and most widely deployed network technology in the world.

Ethernet ports enable you not only to create a small home network, but also to connect to the Internet via a Digital Subscriber Line (DSL) or modem, which requires an Ethernet connection. Ethernet is a shared-media LAN, which means that all stations on the segment use a part of the total bandwidth. Depending on the type of Ethernet implemented, this total bandwidth is a 10 Mbps (Ethernet), 100 Mbps (Fast Ethernet), or 1000 Mbps (Gigabit Ethernet).

The next component that you will need to setup a network is a hub or sometimes a switch. A hub is a device that connects the cables from computers and other devices such as printers in a network. Hub is a multiport repeater and is used to create collision domains, in which all devices on the network can see each other. Hubs provide the signal amplification required to allow a segment to be extended a greater distance. A hub takes an incoming signal from anyone of the ports and repeats on each of its out ports to enable users to share the Ethernet network resources. A switch is a device similar to a hub that enables the connection of multiple computers, access points, and other network enabled devices. The difference between a hub and a switch is that a switch filters the data that passes through it and a hub does not.

The switch is an intelligent device that sends packets to the destined computer only and does not broadcast to all the machines on the network. Ethernet LANs use Media Access Control (MAC) address to determine how traffic is transferred between network segments. Ethernet hubs,

defined by Open System Interconnection (OSI) model physical layer (Layer 1) repeat only the physical signal; the hub does not look at a source or destination address. Ethernet bridges and switches use the source and destination MAC address, defined by the OSI data link layer (Layer 2) to build an interface table and to determine which segment should receive the frame. Routers use the network address, found at the OSI network layer (Layer 3) to build a routing table.

MAC address is a unique serial number assigned and burned into each network adapter that differentiates network cards, just as your house number is unique on your street and identifies your home among others. A MAC address is the physical address of the device. It is 48 bits (6 bytes) long and is made up by two parts: the organizational unique identifier (OUI) and the vendor-assigned address.

The MAC address on a computer might look like this: 00-08-a1-08-c8-13. The OUI is 00-08-a1, and the vendor-assigned number is 08-c8-13.

Computer networks also differ in their design. The two types of high-level network design are called client-server and peer-to-peer. Client-server networks feature centralized server computers that store email, Web pages, files and applications. On a peer-to-peer network, conversely, all computers tend to support the same functions. Client-server networks are much more common in business and peer-to-peer networks much more common in homes.

A network topology represents its layout or structure from the point of view of data flow. In so-called bus networks, for example, all of the computers share and communicate across one common conduit, whereas in a star network, all data flows through one centralized device. Common types of network topologies include bus, star and ring.

In networking, the communication language used by computer devices is called the protocol. Networks often implement multiple protocols to support specific applications. The most common protocol found on the Internet and in home networks is TCP/IP (Transmission Control Protocol/Internet Protocol).

1. **conspicuous** – видимый, заметный
2. **repository** – хранилище
3. **networked units** – объединенные в сеть единицы
4. **to span** – располагаться
5. **wide area network** – глобальная компьютерная сеть
6. **dedicated line** – выделенная линия
7. **to lease** – брать или сдавать в аренду
8. **cabling** – монтаж кабельной проводки, система кабелей
9. **Digital Subscriber Line** – цифровая абонентская линия связи
10. **bandwidth** – полоса пропускания, пропускная способность, скорость передачи информации по каналу
11. **hub** – концентратор
12. **switch** – свитч (сетевой коммутатор)
13. **repeater** – ретранслятор, повторитель
14. **collision domains** – область коллизий, коллизионный домен (Часть сети (сегмент), в котором станции используют общую среду передачи.
При попытке одновременной передачи данных двумя или более станциями возникает конфликт (коллизия))
15. **signal amplification** – усиление сигнала
16. **access point** – точка беспроводного доступа
17. **to broadcast** – транслировать, осуществлять широковещательную передачу
18. **Media Access Control (MAC)** – управление доступом к среде передачи
19. **Open System Interconnection (OSI)** – взаимодействие открытых систем

20. **layer** – слой
21. **frame** – кадр
22. **routing table** – таблица маршрутизации
23. **organizationally unique identifier (OUI)** – уникальный идентификатор организации
24. **vendor-assigned address** – серийный адрес
25. **client-server networks** – сеть типа клиент/сервер
26. **peer-to-peer network** – сеть с равноправными узлами, одноранговая сеть
27. **data flow** – поток данных
28. **network topology** – сетевая топология
29. **conduit** – канал
30. **multiple protocols** – различные протоколы

Wi-Fi

Wi-Fi is a brand originally licensed by the Wi-Fi Alliance to describe the underlying technology of wireless local area networks (WLAN) based on the IEEE 802.11 specifications. It was developed to be used for mobile computing devices, such as laptops, in LANs, but is now increasingly used for more services, including Internet and VoIP phone access, gaming, and basic connectivity of consumer electronics such as televisions and DVD players, or digital cameras.

A typical Wi-Fi setup contains one or more Access Points (APs) and one or more clients. An AP broadcasts its SSID (Service Set Identifier, «Network name») via packets that are called beacons, which are usually broadcast every 100 ms. The beacons are transmitted at 1 Mbit/s, and are of relatively short duration and therefore do not have a significant effect on performance. Since 1 Mbit/s is the lowest rate of Wi-Fi it assures that the client who receives the beacon can communicate at least 1 Mbit/s. Based on the settings (e.g. the SSID), the client may decide whether to connect to an AP. If two APs of the same SSID are in range of the client, the client firmware might use signal strength to decide which of the two APs to make a connection to. Since Wi-Fi transmits in the air, it has the same properties as a non-switched Ethernet network.

Except for 802.11a, which operates at 5 GHz, Wi-Fi uses the spectrum near 2.4 GHz, which is standardized and unlicensed by international agreement; although the exact frequency allocations vary slightly in different parts of the world, as does maximum permitted power.

The frequencies for 802.11 b/g span 2.400 GHz to 2.487 GHz. Each channel is 22 MHz wide yet there is a 5 MHz step to the next higher channel. The maximum numbers of available channels for Wi-Fi enabled devices are: 13 for Europe, 11 for North America, 14 for Japan. In North America, only channels 1, 6, and 11 are deployed for 802.11 b/g.

Wi-Fi allows LANs to be deployed without cabling, typically reducing the costs of network deployment and expansion. Spaces where cables cannot be run, such as outdoor areas and historical buildings, can host wireless LANs. Wi-Fi networks support roaming, in which a mobile client station such as a laptop computer can move from one access point to another as the user moves around a building or area. Wi-Fi products are widely available in the market. Different brands of access points and client network interfaces are interoperable at a basic level of service. However, there are some issues. Wi-Fi signal can be interrupted by other devices, notably 2.4 GHz cordless phones and microwave ovens. Many 2.4 GHz 802.11b and 802.11g Access points default to the same channel, contributing to congestion on certain channels. Wi-Fi networks have limited range. A typical Wi-Fi home router using 802.11b or 802.11g with a stock antenna might have a range of 45 m indoors and 90 m outdoors. Range also varies with frequency band, as Wi-Fi is no exception to the physics of radio wave propagation. Wi-Fi in the 2.4 GHz frequency block has better range than Wi-Fi in the 5 GHz frequency block, and less range than the oldest Wi-Fi (and pre-Wi-Fi) 900 MHz block. Outdoor range with improved antennas can be several kilometers or more with line-of-sight.

Wi-Fi pollution, meaning interference of a closed or encrypted access point with other open access points in the area, especially on the same or neighboring channel, can prevent access and interfere with the use of other open access points by others caused by overlapping channels in the 802.11g/b spectrum as well as with decreased signal-to-noise ratio (SNR) between access points. This is a widespread problem in high-density areas such as large apartment complexes or office buildings with many Wi-Fi access points.

Wi-Fi networks can be monitored and used to read and copy data (including personal information) transmitted over the network when no encryption is used.

Popular Wi-Fi equipment is the next. A wireless access point (AP) connects a group of wireless stations to an adjacent wired local area network (LAN). An access point is similar to an Ethernet hub, but instead of relaying LAN data only to other LAN stations, an access point can relay wireless data to all other compatible wireless devices as well as to a single connected LAN device,

in most cases an Ethernet hub or switch, allowing wireless devices to communicate with any other device on the LAN.

A wireless router integrates a wireless access point with an IP router and an Ethernet switch. The integrated switch connects the integrated access point and the integrated Ethernet router internally, and allows for external wired Ethernet LAN devices to be connected as well as a (usually) single WAN device such as cable modem or DSL modem.

A wireless Ethernet bridge connects a wired network to a wireless network. This is different from an access point in the sense that an access point connects wireless devices to a wired network at the data-link layer. Two wireless bridges may be used to connect two wired networks over a wireless link, useful in situations where a wired connection may be unavailable, such as between two separate homes.

A wireless range extender (or wireless repeater) can increase the range of an existing wireless network by being strategically placed in locations where a wireless signal is sufficiently strong and nearby locations that have poor to no signal strength. An example location would be at the corner of an L-shaped corridor, where the access point is at the end of one leg and a strong signal is desired at the end of the other leg.

1. **underlying technology** – основная технология
2. **VoIP phone access** – телефонный VoIP-доступ
3. **Access Points** – точка доступа
4. **SSID (Service Set Identifier)** – идентификатор беспроводной сети
5. **beacon** – широковещательный идентификатор сети
6. **performance** – производительность
7. **client firmware** – микропрограмма на клиентской части
8. **signal strength** – уровень сигнала
9. **spectrum** – сетка (частот)
10. **expansion** – увеличение размеров, расширение
11. **outdoor areas** – открытая территория
12. **roaming** – роуминг, связь в глобальном масштабе
13. **be interrupted** – прерываться
14. **limited range** – ограниченную дальность
15. **line-of-sight** – зона прямой видимости
16. **Wi-Fi pollution** – Wi-Fi-загрязнение
17. **overlapping channels** – перекрывающиеся каналы
18. **signal-to-noise ratio (SNR)** – соотношение "сигнал-шум"
19. **high-density area** – густонаселённый район
20. **encryption** – шифрование
21. **widespread problem** – широко распространённая проблема
22. **to monitor** – наблюдать, контролировать
23. **to relay** – ретранслировать
24. **compatible** – совместимый
25. **wireless router** – беспроводной маршрутизатор
26. **wireless Ethernet bridge** – беспроводной Ethernet-мост
27. **data-link layer** – уровень канала связи
28. **wireless link** – беспроводная линия связи
29. **wireless range extender** – беспроводной усилитель сигнала
30. **wireless repeater** – беспроводной повторитель

Artificial neural networks and their application

An Artificial Neural Network (ANN) is an information processing paradigm that is inspired by the way biological nervous systems process information. It is composed of a large number of highly interconnected processing elements (neurones) working in unison to solve specific problems. ANNs, like people, learn by example. An ANN is configured for a specific application, such as pattern recognition or data classification, through a learning process. A trained neural network can be thought of as an "expert" in the category of information it has been given to analyse. This expert can then be used to provide projections given new situations of interest and answer "what if" questions. Neural networks take a different approach to problem solving than that of conventional computers. Conventional computers use an algorithmic approach i.e. the computer follows a set of instructions in order to solve a problem. Unless the specific steps that the computer needs to follow are known the computer cannot solve the problem. That restricts the problem solving capability of conventional computers to problems that we already understand and know how to solve. But computers would be so much more useful if they could do things that we don't exactly know how to do.

Neural networks process information in a similar way the human brain does. The network is composed of a large number of highly interconnected processing elements (neurones) working in parallel to solve a specific problem. Neural networks learn by example. They cannot be programmed to perform a specific task. The examples must be selected carefully otherwise useful time is wasted or even worse the network might be functioning incorrectly. The disadvantage is that because the network finds out how to solve the problem by itself, its operation can be unpredictable.

On the other hand, conventional computers use a cognitive approach to problem solving; the way the problem is to solved must be known and stated in instructions. These instructions are then converted to a high-level language program and then into machine code that the computer can understand. These machines are totally predictable; if anything goes wrong is due to a software or hardware fault.

An artificial neuron is a device with many inputs x(n) and one output. The neuron has two modes of operation; the training mode and the using mode. In the training mode, the neuron can be trained to fire (or not), for particular input patterns. In the using mode, when a taught input pattern is detected at the input, its associated output becomes the current output. If the input pattern does not belong in the taught list of input patterns, the firing rule is used to determine whether to fire or not. A firing rule determines how one calculates whether a neuron should fire for any input pattern. It relates to all the input patterns, not only the ones on which the node was trained.

Each input is multiplied by its corresponding weights. Weights are the information used by the neural network to solve a problem. Typically weight represents the strength of the interconnection between neurons inside the Neural Network. The weighted inputs are all summed up inside computing unit (artificial neuron). In case the weighted sum is zero, bias is added to make the output not-zero or to scale up the system response. Bias has the weight and input always equal to ‘1’.

The sum corresponds to any numerical value ranging from 0 to infinity. To limit the response to arrive at the desired value, the threshold value is set up. For this, the sum is passed through activation function.

The activation function is set to the transfer function used to get the desired output. There are linear as well as the nonlinear activation function. Some of the commonly used activation function is - binary, sigmoidal (linear) and tan hyperbolic sigmoidal functions (nonlinear). For Binary activation function, the output has only two values either 0 and 1. For this, the threshold

value is set up. If the net weighted input is greater than 1, an output is assumed one otherwise zero. Sigmoidal Hyperbolic function has 'S' shaped curve. Here tan hyperbolic function is used to approximate output from net input. The function is defined as - f (x) = (1/1+ exp(-σx)) where σ - steepness parameter.

ANNs have the ability to learn and model non-linear and complex relationships, which is important because in real-life, many of the relationships between inputs and outputs are non-linear as well as complex. ANNs can generalize — after learning from the initial inputs and their relationships, it can infer unseen relationships on unseen data as well, thus making the model generalize and predict on unseen data. Unlike many other prediction techniques, ANN does not impose any restrictions on the input variables. Additionally, many studies have shown that ANNs can better model heteroskedasticity i.e. data with high volatility and non-constant variance, given its ability to learn hidden relationships in the data without imposing any fixed relationships in the data. This is something very useful in financial time series forecasting (e.g. stock prices, sales forecasting, industrial process control, data validation).

ANN are also used in the following specific paradigms: recognition of speakers in communications; diagnosis of hepatitis; recovery of telecommunications from faulty software; interpretation of multimeaning Chinese words; undersea mine detection; texture analysis; three-dimensional object recognition; hand-written word recognition; and facial recognition. ANNs are playing a big role in image and character recognition. Character recognition like handwriting has lot of applications in fraud detection (e.g. bank fraud) and even national security assessments. Image recognition is an ever-growing field with widespread applications from facial recognition in social media, cancer detention in medicine to satellite imagery processing for agricultural and defense usage. The research on ANN now has paved the way for deep neural networks that forms the basis of "deep learning" and which has now opened up all the exciting and transformational innovations in computer vision, speech recognition, natural language processing — famous examples being self-driving cars

1. **to inspire** – вдохновить, инспирировать
2. **to process information** – обрабатывать информацию
3. **interconnected** – соединённые друг с другом, взаимосвязанный
4. **to work in unison** – работать слаженно
5. **pattern recognition** – распознавание образов
6. **projection** – видение ситуации
7. **algorithmic approach** – алгоритмический подход
8. **to restrict** – ограничивать, держать в определённых пределах
9. **solving capability** – разрешающая способность
10. **training mode** – режим обучения
11. **using mode** – режим работы
12. **firing rule** – правило срабатывания
13. **to determine** – установить, определить
14. **to relate** –устанавливать отношение
15. **node** – узел
16. **weight** – вес
17. **bias** – смещение
18. **threshold** – порог
19. **activation function** – функция активации
20. **hyperbolic sigmoidal function** – гиперболическая сигмоидальная функция
21. **steepness parameter** – параметр крутизны
22. **non-linear relationship** – нелинейные связи

23. **to generalize** – делать общие выводы
24. **to infer** – предполагать
25. **to impose** –вводить, накладывать
26. **heteroskedasticity** – гетероскедастичность, неоднородность наблюдений
27. **forecasting** – прогнозирование
28. **facial recognition** – распознавание лиц
29. **deep learning** – глубинное обучение
30. **self-driving cars** – беспилотный автомобиль

RFID and NFC technologies

Long checkout lines at the grocery store are one of the biggest complaints about the shopping experience. Soon, these lines could disappear when the ubiquitous Universal Product Code (UPC) bar code is replaced by smart labels, also called radio frequency identification (RFID) tags.

Imagine going to the grocery store, filling up your cart and walking right out the door. No longer will you have to wait as someone rings up each item in your cart one at a time. Instead, these RFID tags will communicate with an electronic reader that will detect every item in the cart and ring each up almost instantly. The reader will be connected to a large network that will send information on your products to the retailer and product manufacturers. Your bank will then be notified and the amount of the bill will be deducted from your account. No lines, no waiting.

RFID technology has been around since 1970, but until recently, it has been too expensive to use on a large scale. RFID tags, a technology once limited to tracking cattle, are tracking consumer products worldwide. Outside the realm of retail merchandise, RFID tags are tracking vehicles, railroad cars and airline luggage, airline passengers, Alzheimer's patients and pets.

Newer innovations in the RFID industry include active, semi-active and passive RFID tags. These tags can store up to 2 kilobytes of data and are composed of a microchip, antenna and, in the case of active and semi-passive tags, a battery. The tag's components are enclosed within plastic, silicon or sometimes glass.

At a basic level, each tag works in the same way. Data stored within an RFID tag's microchip waits to be read. The tag's antenna receives electromagnetic energy from an RFID reader's antenna. Using power from its internal battery or power harvested from the reader's electromagnetic field, the tag sends radio waves back to the reader. The reader picks up the tag's radio waves and interprets the frequencies as meaningful data.

Active and semi-passive RFID tags use internal batteries to power their circuits. An active tag also uses its battery to broadcast radio waves to a reader, whereas a semi-passive tag relies on the reader to supply its power for broadcasting. Because these tags contain more hardware than passive RFID tags, they are more expensive. Active and semi-passive tags are reserved for costly items that are read over greater distances – they broadcast high frequencies from 850 to 950 MHz that can be read 100 feet (30.5 meters) or more away. If it is necessary to read the tags from even farther away, additional batteries can boost a tag's range to over 300 feet (100 meters).

Passive RFID tags rely entirely on the reader as their power source. These tags are read up to 20 feet (six meters) away, and they have lower production costs, meaning that they can be applied to less expensive merchandise. These tags are manufactured to be disposable, along with the disposable consumer goods on which they are placed. Whereas a railway car would have an active RFID tag, a bottle of shampoo would have a passive tag.

Another factor that influences the cost of RFID tags is data storage. There are three storage types: read-write, read-only and WORM (write once, read many). A read-write tag's data can be added to or overwritten. Read-only tags contain only the data that is stored in them when they were made. WORM tags can have additional data (like another serial number) added once, but they cannot be overwritten.

NFC, or Near Field Communication, is a short range wireless RFID communication technology. It means that two devices equipped with NFC technology are able to communicate with each other and share information as soon as they are close to one another. So, they create a "near field" (maximum approximately 10 centimeters) and use high frequencies that will allow to interact. NFC technology is promising because it presents the next evolution of convenient payment with an added layer of security. Some credit cards have NFC chips embedded in them and can be tapped against NFC payment terminals instead of swiped, which eliminates the possibility that someone could skim your data via the magnetic strip. This same system works with cellular phones, too.

As a contactless technology, NFC is one of the many communication tools used to give instant access to any digital data and information from your mobile device. There are 3 different ways to do this. First, the host card emulation mode. This is the "passive" mode, by which a mobile terminal behaves like a smart contactless chip. The phone sends the information to the NFC device, e.g. to a ticket validation kiosk or a payment terminal. It can serve various purposes: mobile payment, show or transportation tickets, coupons, access control, etc. Second, the reader mode. In this case, the mobile has an "active" role as a contactless chip reader. This mode allows you to read information or trigger actions by simply "tapping" or "bumping" your phone close to a NFC tag. NFC tags can be placed on posters, billboards, monuments, bus stops or products. The focus here is to provide a seamless user experience across channels with a specific digital content, from offline to mobile. Third, the peer-to-peer mode. In this mode, the data exchange works in both directions between two mobile devices communicating with each other via NFC. Common applications are to exchange contact information (vCard) between two smartphones, to transfer files and instantly execute money transfers as well.
RFID is a technology allowing to identify and characterise an object thanks to a sticker broadcasting radio waves. It is widely used for logistical purposes in many different sectors where tracking merchandises is a key issue. NFC is derived from RFID but is different in many ways. One of the main differences is the fact that NFC allows a limited data sharing between the tag and the reader. Then, another one is that NFC has a really short range compared to RFID

1. **ubiquitous** – повсеместно распространённый
2. **Universal Product Code** – универсальный товарный код
3. **label** – ярлык, этикетка, бирка
4. **radio frequency identification (RFID) tags** – метки радиочастотной идентификации
5. **electronic reader** – электронное считывающее устройство
6. **retailer** – розничный торговец
7. **manufacturer** – изготовитель, производитель
8. **bill** – документ на оплату
10. **realm** – сфера, область
11. **retail merchandise** –розничные товары, розничная торговля
12. **semi-active** – полуактивный
13. **to harvest** – получать, собирать
14. **radio waves** – радиоволны
15. **to boost** – увеличивать, усиливать
16. **tag's range** – радиус действия метки
17. **disposable** – одноразового использования
18. **Near Field Communication (NFC)** – беспроводная связь ближнего радиуса действия
19. **high frequencies** –высокая частота (3-30 МГц)
20. **to skim** – снимать, считать
21. **magnetic strip** – магнитная полоса
22. **host card emulation mode** – режим эмуляции карты (устройство NFC ведет себя как существующая бесконтактная карта)

23. **seamless** – беспроблемный, гладкий, плавный
24. **instant access** – мгновенный доступ
25. **contactless chip** – бесконтактный чип
26. **reader mode** – режим считывания (устройство NFC является активным и считывает пассивную RFID-метку, например для интерактивной рекламы)

27. **to trigger actions** – инициировать, запустить действие
28. **peer-to-peer mode** – пиринговый режим (два устройства NFC вместе связываются и обмениваются информацией)
29. **money transfer** – перевод денежных средств
30. **data sharing** – совместный доступ к данным

Read and translate the text

What are logic gates?[1]

A logic gate is a device that acts as a building block for digital circuits. Logic gates perform basic logical functions fundamental to digital circuits. Most electronic devices we use today have some form of logic gate in them.

Logic gates can be used, for example, in digital electronics such as smartphones and tablets or in memory devices. Commonly used in integrated circuits (IC), logic gates are also crucial building blocks for electrical and electronic components like counters, processors, resistors, transistors and diodes.

How do logic gates work?

In a circuit, a logic gate works based on a combination of digital signals from its inputs. Most logic gates have two inputs and one output. Applying Boolean algebra, the gate performs some logical operation on the inputs to make a decision and produce a single binary output. At any moment, every terminal is in one of the two binary conditions: true (high) or false (low). False is represented as 0 and true as 1.

Think of a logic gate like a light switch, where in one position, the output is off (0), and another, it's on (1). Depending on the type of logic gate used and the combination of inputs, the binary output will differ. In an AND gate, the output will be 1 if both its inputs are 1. An OR gate will produce an output of 1 if only one of the inputs is 1. Combining multiple logical gates in an IC can create different combinations of outputs for different combinations of inputs.

Basic logic gates

There are seven basic logic gates: AND, OR, XOR, NOT, NAND, NOR and XNOR.

AND gate

The AND gate produces an output of true only when both inputs are true. Otherwise, the output is false. In other words, the output is 1 only when both inputs are 1, and 0 if even one of the inputs is 0. The gate is named so because, if 0 is false and 1 is true, the gate acts in the same way as the logical *and* operator. Figure 1 provides a truth table showing the circuit symbol and logic combinations for an AND gate. In the symbol, the input terminals are on the left, and the output terminal is on the right.



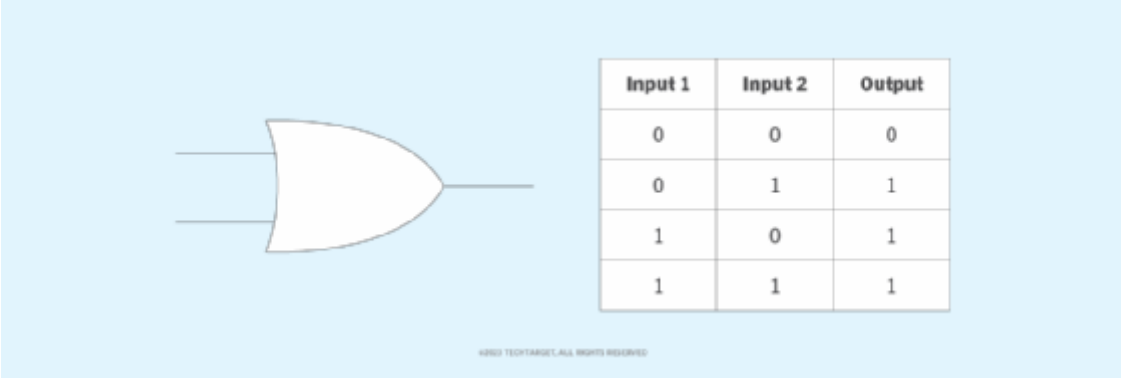
Input 1	Input 2	Output
0	0	0
0	1	0
1	0	0
1	1	1

Figure 1. A diagram of the AND logic gate.

OR gate

In the OR gate, which is among the most widely used logic circuits in digital devices and components, the output is true if one or both inputs are true. If both inputs are false, the output is false. For the output to be 1, at least one input must be 1. The gate gets its name from behaving like the logical inclusive *or*. Figure 2 shows a truth table for the OR gate.

[1] https://www.techtarget.com/whatis/definition/logic-gate-AND-OR-XOR-NOT-NAND-NOR-and-XNOR

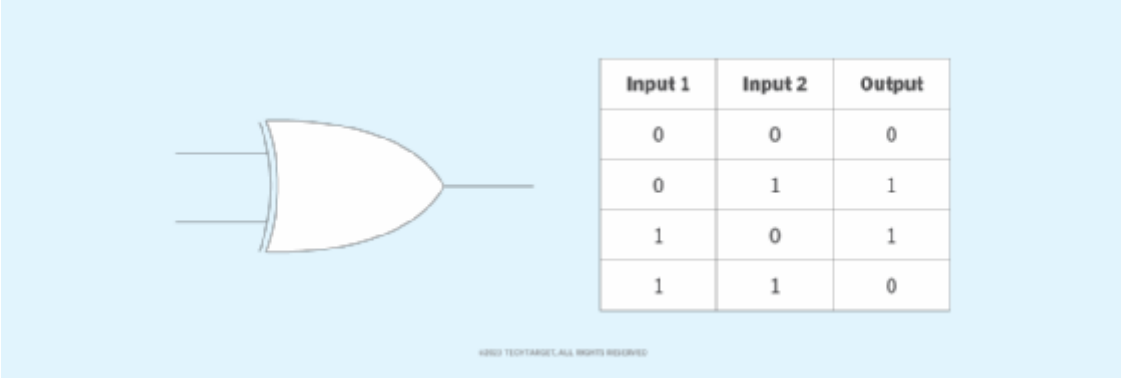


Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	1

Figure 2. A diagram of the OR logic gate.

XOR gate

The XOR, or exclusive OR, gate acts in the same way as the logical *either/or*. The output is true if either input is true, but not both. The output is false if both inputs are false or both inputs are true. Similarly, the output is 1 (true or high) if the inputs are different, but 0 (false or low) if the inputs are the same. Figure 3 shows a truth table for a 2-input XOR gate.

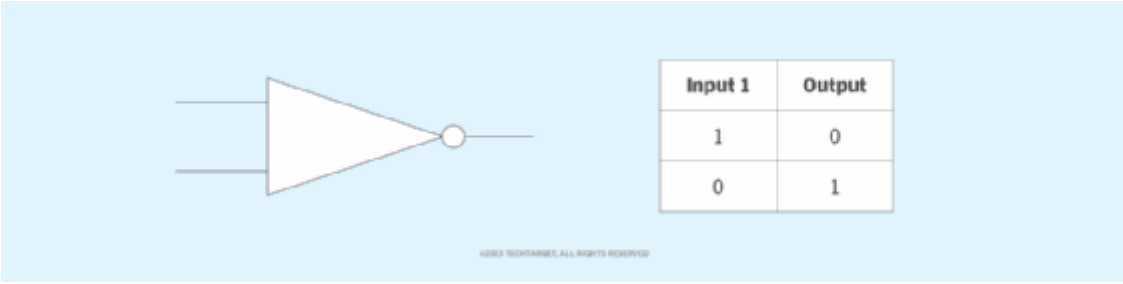


Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	0

Figure 3. A diagram of the XOR logic gate.

NOT gate

A logical inverter, sometimes called a NOT gate to differentiate it from other types of electronic inverter devices, has only one input. A NOT gate reverses the logic state. If the gate's input is 1, then the output is 0. If the input is 0, the output is 1. One of the only logic gates that accepts a single input and produces a single output, the NOT gate is also one of the simplest since all it does is reverse the input applied. As shown in Figure 4, a truth table for a NOT gate is simple.



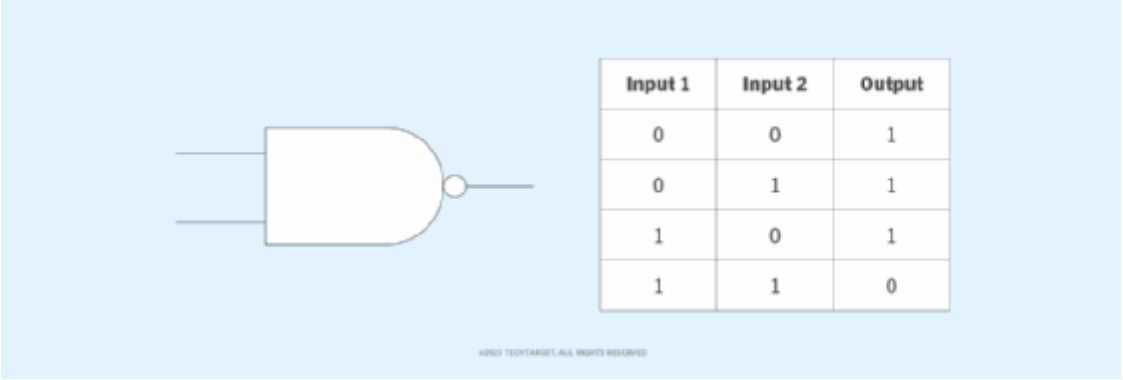
Input 1	Output
1	0
0	1

Figure 4. A diagram of the inverter or NOT logic gate.

NAND gate

The negated AND, or NAND, gate operates as an AND gate followed by a NOT gate. Its symbol is an AND gate with the circle of a NOT gate at the output. The NAND gate produces a false or low output if both inputs are true. Otherwise, the output is true. Another way to visualize it is that a NAND gate inverts the output of an AND gate. It acts in the manner of the logical operator *and*

followed by negation. Figure 5 provides a truth table showing the logic of a NAND gate for two types of inputs.

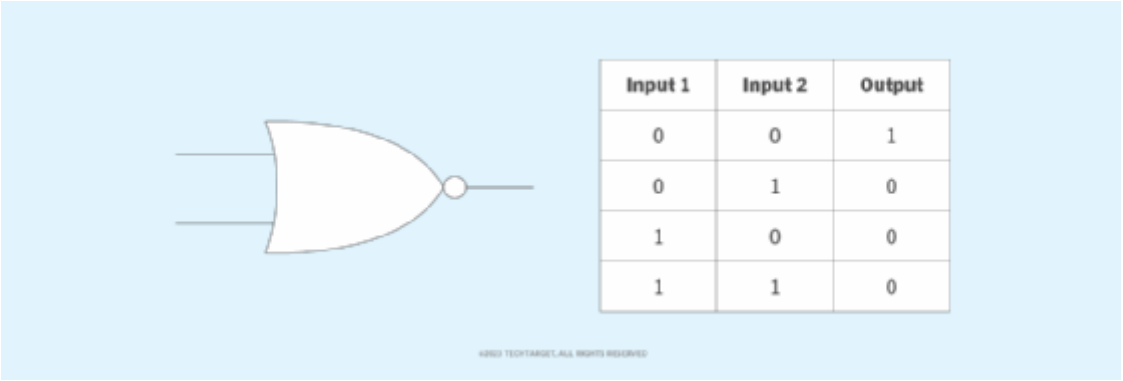


Input 1	Input 2	Output
0	0	1
0	1	1
1	0	1
1	1	0

Figure 5. A diagram of the NAND logic gate.

NOR gate

The NOR, or NOT OR, gate is a combination OR gate followed by an inverter. Its output is true or high if both inputs are false or low (logic 0). Otherwise, the output is false. This logic is easy to understand with the gate's truth table, shown in Figure 6.

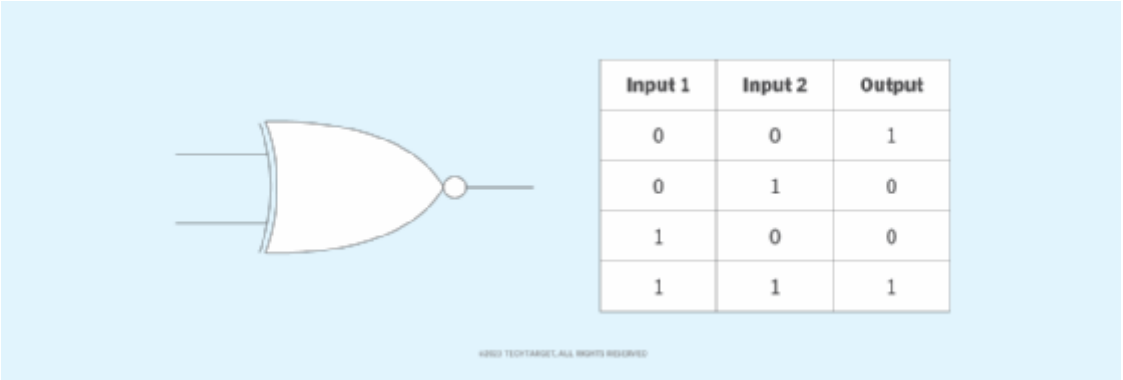


Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	0

Figure 6. A diagram of the NOR logic gate.

XNOR gate

The XNOR, or exclusive NOR, gate is a combination of an XOR gate followed by an inverter. Its output is true if the inputs are the same and false if the inputs are different. It produces a true output, regardless of whether the inputs are true or false. However, they must both be the same for the output to be true. The XNOR truth table looks like Figure 7.



Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	1

Figure 7. A diagram of the XNOR logic gate.

Universal logic gates

The NAND and NOR gates are known as universal logic gates since they can be implemented in digital circuits to perform any Boolean function without additional logic gates. These gates can

also replicate the functionality and behavior of the other gates. In doing so, they help to simplify the design -- and often the cost -- of digital ICs.
For example, two NAND gates can be used to create an AND gate. In this case, the first NAND gate will inverse the logical AND operation of the two inputs. The second NAND gate then inverts this output to produce the original AND logic. Similarly, three NAND gates can be used to create an OR gate. The first NAND gate will inverse the first input, the second will inverse the output of the first NAND gate, and the third NAND gate will produce the logical OR of the outputs of the first two NAND gates.
NAND gates can also be used to simulate the outputs of XOR, NOR and XNOR gates. Similarly, the NOR logic gate can be used to replicate the output of all these logic gates:

- AND gate.
- OR gate.
- NAND gate.
- XNOR gate.
- XOR gate.

Composition of logic gates

High or low binary conditions are represented by different voltage levels. The logic state of a terminal can, and generally does, change as the circuit processes data. In most logic gates, the low state is approximately 0 volts, while the high state is approximately +5 V.
Logic gates can be made of resistors, transistors or diodes. These components are wired together in specific configurations to ensure they transform the inputs in expected ways. Resistors, for example, can commonly be used as a pull-up or pull-down resistor. Pull-up and pull-down resistors are used when there are any unused logic gate inputs to connect to a logic level 1 or 0. This prevents any false switching of the gate.
Likewise, transistors provide switching -- turning on or off in response to input signals -- while diodes ensure current flows in only one direction to stabilize the circuit.
Commonly used logic gates are transistor-to-transistor logic (TTL) and complementary metal-oxide-semiconductor (CMOS). TTL ICs use negative-positive-negative and positive-negative-positive bipolar junction transistors. CMOS ICs are constructed from metal-oxide-semiconductor or junction field-effect transistors. TTL ICs might commonly be labeled as the 7400 series of chips, while CMOS ICs might often be marked as a 4000 series of chips.

COMPRESSION

Answer the following questions:

1. What term is used to describe a device that serves as a fundamental component from which more complex digital circuits are constructed?
2. Apart from smartphones and tablets, what other category of digital electronic devices is explicitly mentioned as containing logic gates?
3. Among the various electrical and electronic components that rely on logic gates as essential building blocks, which four are specifically named in the passage?
4. According to the text, what mathematical framework is applied to the input signals of a gate in order for that gate to perform a logical operation and reach a decision?
5. What are the two binary conditions that any terminal within a circuit can exhibit at any given moment, and which numerical values correspond to each?
6. The text draws an analogy between a logic gate and a common household device. What is that device, and what does its two possible positions represent in terms of binary output?

7. In the case of an AND gate, under what precise combination of input values will the output be 1, and under what condition will it be 0?
8. For an OR gate, what is the minimum number of inputs that must be 1 in order for the output to be 1, and what is the only situation in which the output becomes 0?
9. What does the acronym XOR stand for, and how does the behaviour of an XOR gate differ from that of a standard OR gate when both inputs are true?
10. Apart from "NOT gate," what alternative name is given to this component, and what fundamental operation does it perform on its input signal?
11. How many input terminals does a NOT gate possess, and why is it considered one of the simplest logic gates?
12. The NAND gate is described as a combination of two other gates. Which two gates, and in what order, are combined to form a NAND gate?
13. What distinguishing feature appears on the circuit symbol of a NAND gate that differentiates it from an AND gate symbol?
14. A NAND gate produces a false (low) output under one specific condition. What is that condition?
15. What is the output of a NOR gate when both of its inputs are false (logic 0), and what is its output in all other cases?
16. Which two logic gates are referred to as "universal logic gates," and what property justifies this designation?
17. According to the example provided, how many NAND gates are required to construct an OR gate, and what is the intermediate step involving the first two of those gates?
18. In most logic gates, what approximate voltage corresponds to a low (false) binary state, and what approximate voltage corresponds to a high (true) binary state?
19. What is the purpose of using a pull-up or pull-down resistor on an unused input of a logic gate?
20. What are the two commonly used logic gate families mentioned in the text, and by what series numbers are their respective integrated circuits typically labeled?

Task 1. Read the following statements and decide if they are True/False or Not Given.

1. Not a single contemporary electronic device fails to incorporate at least one form of logic gate somewhere within its design.
2. Neither integrated circuits nor any other electronic components would ever require logic gates, according to the passage.
3. A typical logic gate, unless it is a NOT gate, does not deviate from the pattern of having two input terminals and exactly one output terminal.
4. Without the application of Boolean algebra, the logical operations performed by a gate on its incoming signals would remain entirely unchanged.
5. It is impossible for any terminal within a circuit to simultaneously register as being both in a high state and in a low state at any given moment.
6. The analogy comparing a logic gate to a light switch implies that the output cannot occupy any intermediate state between fully off and fully on.
7. For an AND gate to produce a binary output of 1, it is sufficient that no more than one of its inputs is 0.
8. An OR gate will fail to produce a 1 output only in the single case where neither of its inputs is 1.
9. A high output from an XOR gate is never observed whenever the two binary inputs happen to be different from one another.

10. The alternative designation of the NOT gate as a "logical inverter" stems from its tendency to preserve the original logic state rather than altering it.
11. Provided that at least one of its inputs is connected to ground, a NOT gate can function correctly with two inputs instead of one.
12. Distinguishing the visual symbol of a NAND gate from that of an AND gate would be impossible for anyone who ignores the presence of a small circle at the output.
13. The only circumstance under which a NAND gate yields a low output is when both of its inputs are simultaneously high.
14. A NOR gate's output will be high under all conditions except those in which either input is high, regardless of the state of the other input.
15. The output of an XNOR gate is guaranteed to be true if and only if the two inputs are not in opposite states, irrespective of whether those inputs represent true or false values.
16. Despite being capable of substituting for other gate types, universal logic gates lack the ability to realise arbitrary Boolean functions without assistance from additional gates.
17. Recreating the exact functionality of an AND gate is achievable through the interconnection of precisely two NAND gates arranged in a particular configuration.
18. A detailed, sequential schematic illustrating the construction of an XOR gate using only NOR gates is included in the portion of the text discussing universal logic gates.
19. The low binary condition in the majority of logic gates corresponds to a voltage level that is exactly zero volts, with no tolerance for approximation.
20. The primary function of a pull-up resistor, as described in the text, is to tie an unused logic gate input to a logic 0 level in order to prevent erratic switching behaviour.

SPEAKING AND DISCUSSION

Discuss the following questions with a partner or in a small group.

1. Why do you think the text calls logic gates “building blocks” rather than “complete circuits”?
2. The text says most logic gates have two inputs and one output. Why might a NOT gate be an exception?
3. Explain in your own words why Boolean algebra is necessary for logic gates to work.
4. The light switch analogy compares a logic gate to a switch. What is one limitation of this analogy?
5. Based on the description of the AND gate, what would happen if one input is 1 and the other is 0? Why?
6. The OR gate is described as “inclusive or.” What do you think “exclusive or” (XOR) means, and how does it differ?
7. Why might the XOR gate be useful for circuits that need to detect differences between two signals?
8. The NOT gate is also called an inverter. Why is that name more descriptive of its function?
9. How does the NAND gate relate to the AND gate and the NOT gate?
10. If you only had NAND gates, could you build an AND gate? Explain using the text.
11. The text says three NAND gates can create an OR gate. Why would an engineer do that instead of using a single OR gate?
12. What is the practical advantage of having “universal logic gates” like NAND and NOR?
13. Why does the text mention cost saving when using universal gates?
14. How do voltage levels (0V and +5V) relate to binary states in logic gates?
15. What is the purpose of a pull-up or pull-down resistor on an unused input?
16. Why might an unused input cause “false switching” if not connected properly?
17. Compare TTL and CMOS logic in terms of the types of transistors they use.

18. Why do you think the 7400 series (TTL) and 4000 series (CMOS) became industry standards?
19. The text says “the logic state of a terminal can and generally does change as the circuit processes data.” What does this imply about the speed of logic gates?
20. Based on the composition section, which component (resistor, transistor, or diode) is most critical for switching, and why?

Read and translate the text

Integrated Circuits[2]

Integrated circuits (ICs) also called microelectric chips are used to create a device that can perform certain electrical operations such as signal amplification which is called a transistor from the combination of words of the transfer and resistor. Solid-state electronics is the name given to the field of research into how to make electronic devices out of solid materials.

They are standard ICs available that work as power regulators, amplifiers, simple signal processors, and so on. ICs range from 8-pin to 16-pin packages in which the common package has two rows of pins, A 20-pin DIL package will have 2 rows of ten pins. Bigger integrated circuits or chips are used for complex digital applications such as oriented IC.

What is an Integrated Circuit (IC)?

Integrated Circuit (IC) also called micro electric chip, microchip, or chip. It is a group of electronic circuits placed on a metal plate designed with semiconductor materials.

Integrated circuits are one of the indispensable circuit elements of electronics in many electronic devices that we use in our daily lives. it has its origin in the invention of transistors in 1947 by William B. Shockley and his team at AT&T Bell laboratories. The team found that under the right circumstances, electrons would form a barrier at the surface of certain crystals, and they learned to control the flow of electricity (flow of charges) through the crustal by manipulating the barrier.

An integrated circuit (IC) can consist of thousands or even millions of:

- Transistors
- Resistors
- Capacitors
- Diodes

Definition of Integrated Chip

An IC is the fundamental building block of all modern electronic circuits. ICs are made up of several components such as R, C, L, diodes and transistors as active devices and capacitors and resistors as passive devices, and their interconnections are built up on a thin substrate of semiconductor material typically silicon. An IC can function as an amplifier, oscillator, timer, counter, logic gate, microcontroller, or microprocessor.

One common example of a modern-day IC is a computer processor, which typically contains millions or billions of transistors, capacitors, logic gates, etc., connected together to form a complex digital circuit. Although the processor is an IC, not all ICs are processors. so in this article, we will go through what is ICs, the types of ICs, the pros & cons of ICs, and so on. In simple words, Integrated Circuits are the Integration of active components(transistors) and passive components(resistors, capacitors) into a silicon chip.

Integrated Circuit Design

Depending on its input signal type, an integrated circuit can be classified as an analog integrated circuit, a digital integrated circuit, or a mixed circuit.

[2] https://www.geeksforgeeks.org/electronics-engineering/integrated-circuits/

Digital Integrated Circuits

- They are also called Computer and Logistics circuit.
- ICs are used as computer memories (such as RAM and ROM) and microprocessors.
- The circuit is located either in the on-state area or in the off-state area and is not located in between.
- Examples such as AND gate, OR gate, and Flip Flops.

Analog Integrated Circuits

- Analog Circuits are also known as linear circuits. These analog circuits have continuously variable inputs.
- These inputs are also known as analog signals. The output signal then comes out as a linear function of the input signal.
- It is used in audio-frequency or radio-frequency amplifiers.

Mixed Circuits

It is a combination of analog and digital design principles. They function as digital-to-analog converters, analog-to-digital converters, and clock or timing ICs.

THT (Through Hole Technology)

In this circuit, THT is to be soldered punched cards. These ICs can be easily used by mounting on a breadboard**.**

SMD (Surface Mound Device)

In this circuit, SMD is to be soldered on a circuit board. These ICs are much smaller compared to THD ICs because they are designed for use on circuit boards produced on machines.

Linear integrated circuits (Operational-Amplifier)

The relationship between the input and output of the circuit should be linear.

Classification of ICs

There are two types of integrated circuits:

- Linear integrated circuits (Operational-Amplifier)
- Digital integrated circuits.

Operational Amplifier (Op-Amp)

Op-Amp is a fundamental block of electronic functioning that is used to construct all the electronic circuits. It was first used by John R. Ragazzini in 1947. It is a direct coupled high gain amplifier and it consists of one or more different amplifiers followed by a level translator and output stage. Op-Amp is a versatile device that can amplify AC as well as DC input. It mainly consists of multistage amplifiers which get into a number of amplifier stages interconnected to each other in a complex manner.

Digital integrated circuits

Digital integrated circuits are important component in the modern electronics as it is used to create digital design approach which can maximize circuit density and efficiency. These ICs are mostly used in the Computer memories such as RAM and ROM ,as well as in microprocessors.

Integrated Circuit Construction

At first N-type semiconductor is taken and the photolithography process is used. In this technique a layer of silicon dioxide insulator is coated over it then a thin layer of photo-resistant material is placed over So2 after this masking is done UV radiation is made incident on it whenever the light is incident over the surface of photoresistive-material it becomes hard and there is no chemical reaction takes place.

Next Silicon dioxide is removed with the help of a process called Etching, lastly, the required impurities are added the process is several times to obtain different geometrics required to complete the required chip circuit design, and the internal connection between components is made by passing aluminum vapor and then the chip is enclosed with a protective plastic case.

Finally with the help of thin aluminum leads the connectors are taken out and this is how the standard package of IC is manufactured and the IC used in this particular manufacturing process is Monolithic IC.

Functions of Integrated Circuit

The main function of the integrated circuit is that it can perform high-level tasks such as amplification, signal processing, and digital communications etc., the function of IC is cost reduction, miniaturization, and efficiency & performance enhancement among all other circuits. In terms of Cost reduction, IC provides a cheap alternative to requiring a large amount of semiconductor parts and electrical parts mounted on a circuit board through soldering.

The performance and efficiency enhancement function in IC is made possible by implementing complex circuit networks. For instance, when the sound cards in PCs were just introduced manufacturers decided to build medium-scale integration circuits (MSI) that support system sound factors and applications and the main performance enhancement function is low power optimization in turn results in high efficiency.

Integrated Circuit Features

Given below are the feature of the integrated circuit

- **Construction and Packaging:** The ICs are build using components such as silicon. The small gold and aluminum wires connect the components. These wires and components are molded into a flat block of plastic or ceramic. The solid block helps prevent the chip from overheating and keeps it cool.
- **Size of an IC:** The Size of the IC chip is between 1 square mm and more than 200 mm.
- **Integration of an IC:** The Integrated chips have various chip combined together into one chip. Example of an integrated chips is a microcontroller which integrates a microprocessor, memory, and interface into a single unit.

Advantages of Integrated Circuits

The advantages of integrated circuits include the following.

- Small size.
- Power consumption is less.
- Batch Processing results in cost reduction.
- Improves System Reliability
- Less weight.
- More consistent.
- It operates at high temperatures.
- Better Functional Performance.

Disadvantages of Integrated Circuits

The disadvantages of integrated circuits include the following.

- It is very Delicate.
- The power consumption is limited to 10 watts
- The indicators/coils cannot be made.
- A low-temperature coefficient cannot be achieved.
- It handles a limited amount of power.
- Cannot operate at high voltage operation.

Applications of Integrated Circuits

- **Logic Gate:** The logical gate are constructed using ICs. It generates output based on the input signals.
- **Timer:** The Times ICs are used to produce a timing cycle with 100% or 50% duty cycle.
- **Operational Amplifiers:** The Op amp ICs are used produce high gain voltage amplifier.
- **Voltage Regulators:** The Voltage Regulators ICs are used to produce a constant DC output.

Future of Integrated Circuits

As we know IC is the heart of modern technology as we look into the future several indicators and trends are beginning to emerge first, Miniaturization of ICs is expected to grow which is often referred to as **Moores's law** which states that the size of the transistor on any chip will become half approximately two years once.
Secondly, Integrating ICs with Artificial Intelligence (AI) becoming increasingly prevalent that is AI has the potential ability to revolutionize the operations and functioning of IC and that could lead to improvement in efficiency and performance.
Thirdly, smart devices are interconnected nowadays which enables the rise of the Internet of Things (IoT) which increases the demand for ICs.

COMPREHENSION

Answer the following questions:

1. What does the acronym IC stand for, and what are two other names for it?
2. According to the text, who discovered the transistor and in which year?
3. What are the two main categories of components (active and passive) mentioned in an IC?
4. Which semiconductor material is typically used as the substrate for ICs?
5. Name three functions an IC can perform (e.g., amplifier, oscillator, etc.).
6. What is the difference between THT and SMD in terms of mounting?
7. Which type of IC operates with continuously variable input signals?
8. What are two examples of digital integrated circuits given in the text?
9. Who first used the term Operational Amplifier (Op-Amp) and in which year?
10. What is the first step in the IC construction process described?
11. What is the purpose of the photolithography process in IC manufacturing?
12. How are internal connections between components made during IC construction?
13. What material is used to enclose the chip as a protective case?
14. What is the name of the IC type produced by the manufacturing process described?
15. List three main functions of an integrated circuit according to the text.
16. What size range is given for an IC chip (in square mm)?
17. Provide an example of an integrated chip that combines microprocessor, memory, and interface.
18. What is the maximum power consumption limit mentioned for ICs?
19. Name three applications of ICs listed in the text.
20. What is Moore's law as explained in the "Future of Integrated Circuits" section?

Task 1. Read the following statements and decide if they are True/False or Not Given

1. The term "integrated circuit" was first used by William B. Shockley in 1947.
2. All integrated circuits are designed to perform signal amplification as their primary function.
3. A 20-pin DIL package contains two rows of ten pins each.
4. Silicon is the only semiconductor material used for IC substrates.
5. The field of solid-state electronics focuses exclusively on liquid crystal displays.
6. Integrated circuits can contain both resistors and capacitors on the same substrate.
7. A computer processor is an example of an IC, and every IC is a computer processor.
8. The text mentions that mixed circuits are used exclusively for analog-to-digital conversion.
9. Through Hole Technology (THT) ICs require soldering onto punched cards and can be mounted on a breadboard.

10. Surface Mount Device (SMD) ICs are larger than THT ICs because they are designed for machine production.
11. John R. Ragazzini invented the operational amplifier in 1947 for use only in DC applications.
12. The photolithography process described uses a layer of aluminum dioxide insulator.
13. UV radiation hardens the photo-resistant material, preventing chemical reaction in exposed areas.
14. Etching is used to remove the photo-resistant material after UV exposure.
15. Monolithic ICs are manufactured using a process that adds impurities several times to achieve required geometries.
16. Integrated circuits can perform high-level tasks but cannot be used for digital communications.
17. The integration of multiple chips into one unit is exemplified by a microcontroller that includes microprocessor, memory, and interface.
18. The maximum power consumption of an IC is limited to 10 watts due to the protective plastic case.
19. Low-temperature coefficient can be easily achieved in all integrated circuits.
20. According to Moore's law, the number of transistors on a chip doubles approximately every two years while the size of each transistor halves.

SPEAKING AND DISCUSSION

Discuss the following questions with a partner or in a small group:

1. The text states that ICs are "indispensable circuit elements." Do you agree that no modern electronic device could function without them? Justify your reasoning.
2. Why do you think the invention of the transistor in 1947 is considered the origin of the IC, rather than the IC itself being invented then?
3. The text mentions both "active" and "passive" devices. Based on the description, what distinguishes them, and why is that distinction important for circuit design?
4. Considering that a modern computer processor contains billions of transistors, what engineering challenges might arise beyond simply packing more components onto a chip?
5. The text explains THT and SMD. Which technology do you think is more suitable for mass production of smartphones? Give reasons.
6. Linear (analog) ICs are described as having a linear relationship between input and output. What would happen if that relationship became non-linear in a critical system like medical equipment?
7. The text says Op-Amps can amplify both AC and DC inputs. Why is that versatility significant compared to simpler amplifiers?
8. Photolithography involves UV radiation and etching. What potential limitations does this process have when trying to create nanometer-scale circuits?
9. Why might a manufacturer choose a ceramic package over a plastic one for an IC used in aerospace applications?
10. The text lists "improves system reliability" as an advantage. Explain how integrating many components into a single chip could actually increase reliability compared to discrete components.
11. One disadvantage is that "indicators/coils cannot be made" on an IC. Why would the absence of inductors limit certain types of circuits?
12. The text mentions low power consumption as an advantage but also says power consumption is limited to 10 watts. When would this limit be a serious drawback?

13. How might the trend of integrating ICs with AI (mentioned in the future section) change the way ICs are designed compared to traditional logic circuits?
14. The Internet of Things (IoT) increases demand for ICs. What specific properties would IoT devices require from ICs that are different from, say, desktop computer processors?
15. Moore's law states transistor size halves every two years. What physical or economic factors might eventually make this law impossible to sustain?
16. The text separates digital and analog ICs, but mixed-signal ICs exist. Why would a device like a smartphone need mixed-signal ICs rather than purely digital ones?
17. The text states that ICs "cannot operate at high voltage." Why might that be a problem for electric vehicle power management systems?
18. Do you think the advantages of ICs (small size, low weight, low cost) outweigh the disadvantage of being "very delicate" in most consumer applications? Explain.
19. The text describes operational amplifiers as a "fundamental block" of electronic functioning. What would modern signal processing look like without op-amps?
20. Given that ICs are used in timers and voltage regulators, how might a failure in a single voltage regulator IC affect an entire computer system?

Read and translate the text.

BIOS Full Form - Basic Input/Output System[3]

BIOS stands for **Basic Input/Output System.** BIOS is the first program to run when the computer starts. It is a ROM chip (EEPROM chip) found on motherboards and is used for hardware initialization during the booting process. It is pre-installed in the computer system. When BIOS boots up the computer system, it checks if all of the components are in place and working, and then it loads the operating system into the computer's RAM from the hard disk of the system. Nowadays BIOS is written on flash memory, to ease the process of rewriting the BIOS chip, whenever needed.

History of BIOS

BIOS was first introduced in 1975, in the CP/M Operating System. Control Program for Microcomputers, was developed for Intel 8080/85-based microcomputers by Gary Kildall of Digital Research, Inc. The system had a simple bootloader in its ROM. IBM classified the system bios into two categories when PS/2 machines were introduced:

- **CBIOS:** CBIOS stands for **Compatibility BIOS**. The real-mode portion provided backward compatibility with current operating systems and thus was named CBIOS.
- **ABIOS:** ABIOS stands for **Advanced BIOS**. It provided new interfaces, especially the one suited for multitasking OS.

Functions of BIOS

- **POST:** Power on Self-Test is the process performed by BIOS immediately after the computer system starts to check any hardware-related issues. if any error is found in this process, the system alerts through flashing LEDs, audible beeps, or error messages on the monitor, all of which are technically referred to as POST codes, beep codes, and on-screen POST error messages, respectively.
- **Bootstrap Loader:** The bootstrap program stored in ROM is loaded in the hard drive, as soon as the computer starts. The bootstrap loader locates the operating system and passes the control to it.
- **Bootstrap Driver:** Bootstrap Driver configures basic hardware so that the system can run. They are low-level drivers and give basic information about computer hardware.

[3] https://www.geeksforgeeks.org/operating-systems/bios-full-form/

- **CMOS Setup:** Complementary Metal-Oxide Semiconductor (CMOS) chip, saves some system information, such as the time, date, and other essential system settings. CMOS chip is powered by the battery located in the motherboard. It also stores some information about hardware. The chip's content is copied to the RAM(main memory) of the system to increase the speed since the CMOS chip is quite slow. It is because the BIOS and CMOS, setup configure the system's crucial information like the password, date, and time.
- **Boot device selection:** One of the CMOS's most important roles is, it can alter the device boot process. This is important for system restoration because the CMOS may need to change boot priority from the hard drive to the optical drive or flash drive to launch the operating system installer or adjust which hard drive to load the operating system from.

Working of BIOS

- A **Basic Input/Output System (BIOS)** is a kind of program that can be accessed by the EPROM (Erasable Programmable Read Only Memory) chip. Upon switching on the system, the control is passed to the BIOS Program in a similar place on EPROM.

The working of BIOS is mentioned below:

- **Power:** Upon turning on the computer, BIOS starts running.
- **Power-On Self-Test (POST):** BIOS will perform some security checks and make sure that all essential components (RAM, CPU, Storage devices, etc.) are working properly. On getting any issue, it usually beeps or give some error message.
- **Identify Boot Device:** Upon getting confirmation that everything is okay, BIOS looks for Bootable device like USB Drive, CD/DVD.
- **Load the Bootloader:** Once the BIOS finds the boot device, it runs small program called bootloader that is responsible for loading the OS.
- **Start the OS:** Now, the bootloader has responsibility of loading the OS into memory and after which you are able to start the computer.
- **BIOS Setup Utility:** BIOS Setup Utility can be accessed (keys like F2, F10, or Delete during start) that helps in changing boot order and system time.
- **Ongoing Hardware Communication:** Throughout the operation of computer, BIOS helps OS to communicate with the hardware and make sure everything runs smoothly.

Features of BIOS

- Every BIOS implementation is specifically designed for a particular type of motherboard model or particular computer.
- Originally, BIOS firmware was stored in the ROM memory but nowadays it is stored in Flash memory so that it can be rewritten without removing the chip from the motherboard.
- The other reason for doing this is that it allows easy updates of BIOS firmware, so new features can be added and bugs can be fixed.
- The latest advancement of BIOS is UEFI(Unified Extensible Firmware Interface).

Accessing BIOS

With BIOS, many computer applications are not having any restrictions to keep the details of the hardware, attached I/O device, etc. Consumers have access to BIOS and also the configuration of BIOS. Some of the steps mentioned below help in setting up the BIOS.

How to Setup BIOS?

Step 1: First turn off your computer or simply restart it.
Step 2: When the system turns On, wait for the message similar to 'entering setup' to come.
Step 3: Wait for the message 'Press Key to enter BIOS Setup'. Some keys are like Del, Tab, etc.
Step 4: Press the specified key as early as possible.
Step 5: Process Complete.

BIOS Security

BIOS Security is an advanced feature of cybersecurity. It has the work to protect the Operating System from suspicious activities. Plundervolt is one of the manipulations of BIOS which is used to manage computer power supply at the time of memory writing, that has the work to maintain the security gaps which overall results in improving the security of the BIOS.

BIOS Manufacturers

IBM was the first organization that single-handedly owed the BIOS. In spite of IBM, there were several other companies like Phoenix Technologies have their own BIOS. There are several companies that are producing motherboards with embedded BIOS chips in them. Some of the companies are HP, Ricoh, etc. Each organization has its own drivers of motherboards, and each of them creates it in a unique way.

Conclusion

BIOS (Basic Input/Output System) is a crucial component that acts as the bridge between a computer's hardware and its operating system. It initializes and tests hardware during startup, loads the operating system, and allows users to configure basic system settings. Understanding how BIOS works helps users troubleshoot issues and better manage their hardware.

COMPREHENSION

Answer the following questions:

1. What does BIOS stand for and where is it physically located on a computer?
2. What type of memory chip is BIOS typically stored on in modern systems?
3. In which year and on which operating system was BIOS first introduced?
4. Who developed the CP/M operating system and for which microprocessor architecture?
5. What are the two categories of BIOS introduced by IBM with PS/2 machines?
6. What is the full form of CBIOS and what was its main purpose?
7. What does ABIOS stand for and what kind of interface did it provide?
8. List the four main functions of BIOS mentioned in the text.
9. What does POST stand for and what happens if an error is found during POST?
10. What are the three types of error signals mentioned for POST failures?
11. What is the role of the bootstrap loader?
12. What is the difference between a bootstrap loader and a bootstrap driver?
13. What does CMOS stand for and what information does it store?
14. Why is the content of the CMOS chip copied to RAM?
15. How is the CMOS chip powered when the computer is turned off?
16. What role does CMOS play in boot device selection?
17. According to the “Working of BIOS” section, what chip is used to access the BIOS program?
18. What keys are commonly mentioned to access the BIOS Setup Utility?
19. What is the latest advancement of BIOS mentioned in the text?
20. What is “Plundervolt” in the context of BIOS security?

Task 1. Read the following statements and decide if they are True/False or Not Given.

1. BIOS is stored on a ROM chip that cannot be modified under any circumstances in modern computers.
2. The CP/M operating system was developed by IBM for their PS/2 machines.
3. ABIOS was designed to provide interfaces suitable for multitasking operating systems.
4. Power-on Self-Test (POST) errors can only be indicated through audible beeps, not visual signals.

5. The bootstrap loader is responsible for locating the operating system and transferring control to it.
6. CMOS Setup stores system information such as time, date, and password, but not hardware details.
7. The CMOS chip is copied to RAM because the CMOS chip is faster than RAM.
8. The battery located on the motherboard powers the CMOS chip only when the computer is running.
9. Boot device selection can be modified through CMOS to prioritize an optical drive over a hard drive.
10. BIOS can be accessed and configured only by professional system administrators.
11. The BIOS program is accessed by the EPROM chip upon switching on the system.
12. During the boot process, BIOS first identifies the boot device, then runs POST.
13. The BIOS Setup Utility allows users to change the boot order but not the system time.
14. Originally BIOS firmware was stored in flash memory to prevent accidental rewriting.
15. UEFI is a type of BIOS that has been completely replaced by legacy BIOS in modern systems.
16. Plundervolt is a security feature designed to protect BIOS from suspicious activities.
17. IBM was the first organization that exclusively owned BIOS without any competitors.
18. Phoenix Technologies was one of the companies that developed their own BIOS independent of IBM.
19. Each motherboard manufacturer creates BIOS drivers in an identical, standardized way.
20. Understanding BIOS is irrelevant for troubleshooting modern computer hardware issues.

SPEAKING AND DISCUSSION

Discuss the following questions with a partner or in small group.

1. The text states that BIOS is the “first program to run when the computer starts.” What would happen if the BIOS chip became corrupted or its contents were erased? Could the computer still start in any way?
2. BIOS was originally stored in ROM and later moved to flash memory. Why would rewriting the BIOS chip “whenever needed” be both an advantage and a potential security risk?
3. The text mentions that BIOS is “specifically designed for a particular type of motherboard.” Why would a universal BIOS that works on all motherboards be impractical or impossible to create?
4. POST uses beep codes and flashing LEDs to indicate hardware problems. Why do you think modern systems still rely on audible beeps rather than displaying a detailed message on the screen when critical hardware fails?
5. The bootstrap loader locates the operating system and passes control to it. What might happen if the bootstrap loader found multiple bootable devices? How does the system resolve this?
6. CMOS is described as “quite slow,” so its content is copied to RAM. What would be the consequence if a system designer decided to remove the CMOS chip entirely and store its settings directly in the BIOS flash memory?
7. The text separates bootstrap loader and bootstrap driver as two different functions. In practice, why might these two be combined into a single component in some system designs?
8. BIOS Setup Utility can be accessed by pressing keys like Del, F2, or F10 during startup. Why is the timing of pressing the key described as “as early as possible”? What could go wrong if you pressed it too late?

9. The text says BIOS helps the OS communicate with hardware “throughout the operation of computer.” However, modern operating systems often bypass BIOS to talk directly to hardware. Why would they do that?
10. UEFI is mentioned as the latest advancement of BIOS. Based on the limitations described in the text (e.g., CMOS speed, 16-bit real-mode limitations), what improvements would you expect UEFI to offer over legacy BIOS?
11. The text states that “every BIOS implementation is specifically designed for a particular type of motherboard.” How does this affect the ability to upgrade or replace a motherboard without reinstalling the operating system?
12. BIOS Security includes protection from manipulations like Plundervolt. Why would an attack that manipulates power supply during memory writing be particularly difficult to detect?
13. The text lists HP and Ricoh as companies producing motherboards with embedded BIOS chips. Why might a printer manufacturer like Ricoh need its own BIOS implementation rather than using a standard PC BIOS?
14. IBM “single-handedly owed the BIOS” initially. How did the emergence of third-party BIOS manufacturers like Phoenix Technologies change the PC industry?
15. The CMOS battery powers the chip when the system is off. What would happen if the CMOS battery died completely? Would the computer still boot?
16. The text mentions that CMOS can alter the boot device order. Why would changing boot priority from hard drive to USB drive be “important for system restoration”?
17. POST checks “all essential components (RAM, CPU, Storage devices, etc.).” Why might a failed storage device still allow the computer to show error messages, whereas a failed CPU might result in no output at all?
18. The text describes BIOS as a “bridge between hardware and operating system.” In a virtual machine environment (e.g., VMware, VirtualBox), does the virtual machine need its own BIOS? Why or why not?
19. BIOS setup allows changing system time and date. Why does the operating system also maintain its own system time, and how do the two times synchronize?
20. Considering that BIOS is written for 16-bit real-mode processors, how does this legacy affect the boot time of modern computers with 64-bit multi-core processors?

Read and translate the text.

Motherboard[4]

Imagine your computer as a big city, and the motherboard as the main road system connecting all the important buildings—like the CPU (the brain), RAM (the memory), and storage (the library). Without the motherboard, these parts couldn’t talk to each other, and your computer wouldn’t work.

Why Are Motherboards Important?

Motherboards are the backbone of any computer. Here’s why they rock:

- **Connect Everything**: They link all parts, so your CPU, RAM, and storage can work together.
- **Power Supplier**: They send electricity to every component.
- **Upgrade Friendly**: Expansion slots let you add new parts, like a better graphics card for gaming.

Motherboard

How Does a Motherboard Work?

[4] https://www.geeksforgeeks.org/computer-organization-architecture/what-is-a-motherboard/

The power supply transfers electricity to the motherboard so that the computer can use it when you turn it on the computer. Data buses are used to transfer information between the southbridge and northbridge components of the motherboard.
The northbridge motherboard component is used to make the data connections to the CPU, RAM, or PCIe. The RAM starts feeding the CPU inputs. Depending on the type of card which you have, after being written to the PCIe, the data is either copied or relocated to the expansion card. The southbridge motherboard component controls the data connection to the BIOS, the universal serial bus (USB), the serial advanced technology attachment (SATA), and the PCI bus. Now your computer starts because signals are sent to the BIOS.
The left components interact via an electrical signal. A microchip's northbridge or southbridge elements pass through data buses. The data will be encoded into a programming language (1 and 0). When a signal is transferred to a motherboard, the motherboard will start processing and translate information into a language the other component can comprehend.

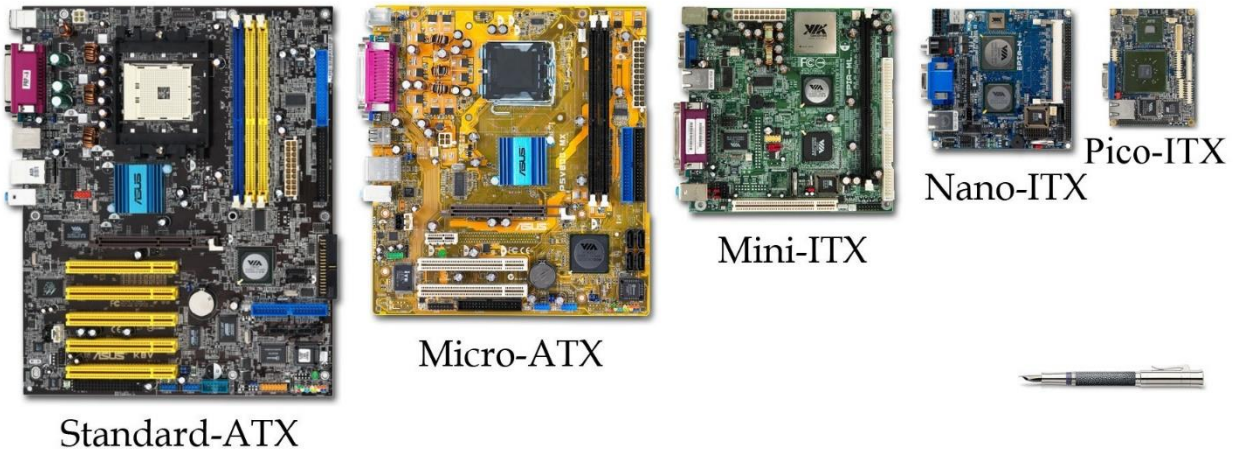
Components of the Motherboard

A motherboard is made of plastic and silicon as well. A motherboard is a complex component with various kinds of ports, slots, and cables connected to it. Some of them are:
It is a hub of a computer system.

- **CPU Slot:** The CPU can be installed in this space. It serves as a connector between a motherboard and a microprocessor. It makes the CPU easier to operate and guards against harm when installing or removing it. It also has a lock to stop the CPU from moving and a heat sink to get rid of the additional heat.
- **RAM Slot:** The RAM (Random Access Memory) is installed in a memory slot or socket on the motherboard. A computer may have two or more memory slots.
- **Expansion Slot:** It is often referred to as an extension port or a bus slot. It is a connection or port on the motherboard that offers a place to put an expansion card for components. It goes by the names bus slot and expansion port as well. It is a port or connection on the motherboard that offers a place to add a hardware expansion card. For instance, you could buy a video expansion card, put it in the expansion slot, and then install a new video card in the computer. AGP, AMR, CNR, PCI, and other popular expansion slots in computers are only a few examples.
- **Capacitor:** Two conducting plates and a thin insulator are placed together to form a capacitor. These components are contained in a plastic bag.
- **Inductor(Coil):** A conducting wire is wrapped around an iron core to create an electromagnetic coil known as an inductor. It stores magnetic energy by functioning as an electromagnet or inductor.
- **Northbridge:** An integrated chip called the northbridge enables communication between the CPU interface, AGP, and memory. Additionally, it enables communication between the southbridge chip and the graphics controller, CPU, and RAM.
- **Southbridge:** Controls the input and output functions. Due to its location, it is referred to as Southbridge. It enables communication between the northbridge chip.
- **USB Port:** You can attach hardware like a mouse and keyboard to your computer using the USB port.
- **PCI Slot:** This term refers to a slot. It enables you to connect Peripheral Component Interconnect Slot (PCI) hardware, like as sound and video cards, network hardware, and modems. These are used to connect the peripherals.
- **AGP Slot:** This is referred to as an Accelerated Graphics Port(AGP) slot. The slot for connecting graphics cards is provided. It is used for displaying graphics on the screen.

- **Heat Sink:** The heat created by the computer processor is absorbed and dispersed by the heat sink. Its main function is to cool down the CPU by absorbing the heat while the system is running.
- **Power Connector:** The purpose of the power connector is to give power to the motherboard.
- **CMOS Battery:** Complementary Metal Oxide Semiconductor is the abbreviation for the CMOS Battery. It is a storage memory. A battery that provides backup power.
- **Integrated Circuits (IC):** IC is also a very important component of the motherboard. Integrated circuit is used for data processing and storage.

Types of Motherboard



There are many types of motherboard as shown below.

1. Advanced Technology (AT) motherboard

These motherboards have sockets and connectors with six prongs apiece for the power connections. Users frequently have difficulties when attempting to connect and use these power connections because it might be challenging to recognise them. Such motherboards were very popular in the 1980s, and production of them lasted well into the 2000s.

2. Standard ATX motherboard

This motherboard comes in 305*244mm (length*breadth) dimensions, these dimensions can vary with different manufacturer. This motherboard offers more expansion slots, up to four slots for RAM, Two or sometimes more than two PCIe slots for dual graphics cards and more USB and other ports for connectivity, Also its size gives space in between components for airflow to keep heat in control.

3. Micro ATX motherboard

This motherboards come in 244*244 mm (length*breadth) dimensions (these dimensions can vary with different manufacturer.). This Motherboard has less ports and slots as compared to Standard ATX board. This type of motherboard is more suitable for those who don't want to much connectivity and later upgrades like adding more ram and additional GPU or Graphics card and adding PCI cards.

4. eXtended ATX motherboard

This motherboard is 344*330 mm dimensions (these dimensions can vary with different manufacturer). This motherboard is designed for both dual CPU and single configuration and has up to 8 RAM slots and has more PCIe and PCI slots for adding PCI cards for different purposes.

5. Flex ATX motherboard

They are regarded as the most compact members of the ATX family. They had a small price tag and were made to take up the least amount of space possible. Mini ATX was modified into Flex ATX by Intel between 1999 and 2000.

6. Low-Profile Extended (LPX) motherboard

In comparison to the previous motherboard, this has two significant changes. The output and input ports were relocated to the device's back as the first change, in comparison to previous iterations, and the second change was to add a riser card, which makes it simpler to attach components.

7. BTX motherboard

A method known as balanced technology extended, or BTX for short, was created to meet the demands of developing technologies, which ask for higher power consumption and, as a result, produce more heat. In order to focus on low-power CPUs, Intel stopped making BTX boards in the middle of the 2000s.

8. Pico BTX motherboard

These boards are referred to as Pico due to their small size when compared to a regular motherboard. Despite the fact that the top half of the BTX is shared, two expansion slots are supported. It is made to satisfy the demands of digital applications, and its distinguishing features include the half-height or riser cards.

9. Mini ITX motherboard

It is significant to note that the information technology extended (ITX) motherboard does not exist in a standard size. The motherboard has been reduced in size and made smaller than in previous generations in its place. It was created in the 2000s, and it is 17 by 17 centimetres in size.

10. Mini STX motherboard

The motherboard presently known as the tiny-STX, which stands for tiny socket technology extended, was originally given the designation "Intel 55". Despite being a 2015 invention, the motherboard is 147 millimetres by 140 millimetres in size.

Tips for Choosing a Motherboard

If you're curious about picking a motherboard (maybe for a school project or PC build), here's what to think about:

- **Match the CPU**: Make sure the motherboard's CPU socket fits your processor (e.g., Intel or AMD).
- **Check RAM Slots**: Get enough slots for your memory needs (2–4 is common).
- **Look at Ports**: More USB ports mean you can plug in more devices.
- **Size Matters**: Pick a size (like Micro ATX) that fits your computer case.

COMPREHENSION

Answer the following questions:

1. According to the city analogy, what does the motherboard represent?
2. List three reasons why motherboards are important according to the text.
3. What component transfers electricity to the motherboard when the computer is turned on?
4. What are data buses used for in the motherboard?
5. Which motherboard component makes data connections to the CPU, RAM, or PCIe?
6. What happens after the RAM starts feeding the CPU inputs?
7. The southbridge controls data connections to which four components or buses?
8. What does the text say happens "when a signal is transferred to a motherboard"?
9. What is the purpose of the lock and heat sink on a CPU slot?
10. What is another name for an expansion slot?
11. Name three popular expansion slots mentioned in the text.
12. How is a capacitor constructed according to the text?
13. What is an inductor made of and what does it store?

14. What communication does the northbridge enable between the southbridge and other components?
15. What kind of hardware can you connect to a PCI slot? Give three examples.
16. What is the primary function of a heat sink?
17. What does CMOS stand for and what is its role on the motherboard?
18. What are integrated circuits (ICs) used for on the motherboard?
19. What are the dimensions of a Standard ATX motherboard according to the text?
20. What are the four tips given for choosing a motherboard?

Task 1. Read the following statements and decide if they are True/False or Not Given.

1. The motherboard is compared to a city's main road system because it controls traffic lights and pedestrian crossings.
2. Motherboards allow users to upgrade components such as graphics cards through expansion slots.
3. The power supply transfers electricity directly to the CPU without involving the motherboard.
4. Data buses are used exclusively for transferring information between the northbridge and the southbridge.
5. The northbridge motherboard component is responsible for making data connections to the CPU, RAM, and PCIe.
6. When data is written to the PCIe, it is always copied to the expansion card without any possibility of relocation.
7. The southbridge controls data connections to the BIOS, USB, SATA, and the PCI bus.
8. Electrical signals on the motherboard are translated into a programming language consisting of 0s and 1s.
9. A CPU slot contains a lock to prevent movement and a heat sink to generate additional heat for cold environments.
10. A computer can have no more than two memory slots on any motherboard.
11. AGP, AMR, CNR, and PCI are examples of expansion slots mentioned in the text.
12. A capacitor is constructed by wrapping a conducting wire around an iron core.
13. An inductor stores magnetic energy and functions as an electromagnet.
14. The northbridge enables communication between the southbridge chip and the graphics controller, CPU, and RAM.
15. The southbridge is named for its geographical location south of the equator on the motherboard.
16. The PCI slot allows connection of network hardware, modems, and sound cards.
17. The heat sink's main function is to generate heat to warm the CPU during winter months.
18. The CMOS battery provides backup power to the Complementary Metal Oxide Semiconductor.
19. Integrated circuits on the motherboard are used only for power distribution, not for data processing.
20. The Micro ATX motherboard has dimensions of 305×244 mm according to the text.

SPEAKING AND DISCUSSION

Discuss the following questions with a partner or in small group.

1. The text describes the motherboard as "the main road system connecting all the important buildings." What limitations does this analogy have when explaining data bus speed differences between northbridge and southbridge?

2. Why would a manufacturer choose to place the northbridge closer to the CPU than the southbridge on the physical motherboard?
3. The text states that data is encoded into a programming language of 1s and 0s before being translated for other components. What would happen if two components used different encoding schemes or voltages?
4. Expansion slots allow adding components like sound or video cards. Why might a user prefer a built-in component on the motherboard over an expansion card, and vice versa?
5. Capacitors and inductors are described as basic components. Why are these rarely discussed in beginner computer hardware guides compared to CPUs or RAM?
6. The northbridge handles communication with the CPU, RAM, and PCIe, while the southbridge handles USB, SATA, and BIOS. What performance problem might arise if a USB device needed to communicate directly with RAM?
7. The text mentions that the CPU slot has a lock to prevent movement and a heat sink to remove heat. Why would movement of the CPU during operation be catastrophic?
8. Some motherboards have multiple PCIe slots for dual graphics cards. What challenges does this create for power supply and cooling that the text does not address?
9. The text lists 10 types of motherboards (AT, ATX, Micro ATX, etc.). Why do you think the original AT motherboard, popular in the 1980s, persisted in production until the 2000s despite its known power connector difficulties?
10. The Flex ATX motherboard was designed to take up minimal space and had a small price tag. What trade-offs in expandability would you expect compared to a Standard ATX board?
11. Intel stopped making BTX boards in the mid-2000s because they focused on low-power CPUs. Why would lower power consumption reduce the need for the BTX thermal design?
12. The Mini ITX motherboard measures 17×17 cm. How might this size constraint affect the placement and performance of the northbridge and southbridge chips?
13. The text says SSDs are “more expensive per unit of capacity than HDDs,” which slowed their adoption. Why might an IT manager still choose SSDs for a critical database server despite the higher cost?
14. A disk array controller implements hardware RAID. What would happen to data on a RAID 0 array if one hard disk failed? How does the text’s definition of RAID (performance, reliability, volume size) apply?
15. The text mentions that firmware is “between hardware and software.” Why is the BIOS considered firmware rather than regular software?
16. The power supply converts AC to DC power. Why would using the wrong voltage (e.g., 220V instead of 110V) damage the motherboard even if the connector fits?
17. Blu-ray discs use a blue laser instead of the red laser used in CDs and DVDs. Why does a shorter wavelength (blue) allow more data storage on the same physical disc size?
18. The text states that floppy disks are “unreasonable to use” because of price per storage compared to flash drives. In what niche scenario might a floppy disk still be preferable today?

19. The selection tip “Match the CPU” says to ensure the motherboard’s CPU socket fits your processor. Why don’t manufacturers create a universal socket that accepts both Intel and AMD processors?
20. The text mentions that “only 0.2% of all microprocessors sold in 2003 were for personal computers.” What does this imply about the importance of embedded systems in modern hardware education?

Read and translate the text.

What are the Functions of a CPU?[5]

A central processing unit, or CPU. The main part of a computer is system processing. Because it carries out commands, performs calculations, and regulates the data flow inside the computer That's why CPU is also known as the brain of the computer. The CPU interprets and executes instructions from the computer's memory, including arithmetic and logical operations, instructions for moving data around, and instructions for controlling flow. By modifying and processing data held in registers, caches, and main memory, it carries out these processes.

Elements of a CPU

- Control Unit (CU)
- Arithmetic Logic Unit (ALU)
- Registers
- Cache
- Bus Interface Unit (BIU)
- Instruction Decoder
- Clock
- Pipeline
- Floating-Point Unit (FPU)

Functions of a CPU

A computer system's central processing unit (CPU) handles a number of crucial tasks. Its principal function is to complete tasks by processing data and following instructions. The following are a CPU's main functions:

Fetching Instructions: The CPU normally retrieves instructions from the computer's memory in a specific order set by the programme counter. It collects the instructions one at a time and gets them ready for processing.

Decoding Instructions: The control unit of the CPU decodes the instructions after they have been fetched. To ascertain the kind of operation to be carried out and the data involved, it analyses the instructions.

Executing Instructions: After decoding, the CPU carries out the required calculations, data manipulations, or control flow activities to carry out the instructions. The actual processing happens in this area. Addition, subtraction, comparisons, and Boolean operations are only a few of the mathematical and logical operations that the arithmetic logic unit (ALU) is in charge of doing.

Managing Registers: The CPU features a number of registers that are used to store data and interim findings. Operands, addresses, and other information needed for calculations are temporarily stored in these registers. Data transfer between registers and memory is controlled by the CPU.

[5] https://www.geeksforgeeks.org/computer-organization-architecture/what-are-the-functions-of-a-cpu/

Controlling Program Flow: The control portion of the CPU directs the flow of instructions and regulates the order of operations. The programme counter and any conditionals or branching instructions determine the next instruction to be executed. Because of this, the CPU may choose among various execution paths and make decisions.
Handling Interrupts: The CPU manages interrupts, which are signals that momentarily stop a programme from running normally. Both internal and external events might cause interruptions, such as keyboards and timers. In order to respond to the interrupt, the CPU pauses the currently running process, saves its state, and then jumps to an interrupt handler code. The CPU continues the paused job after processing the interrupt.
Managing Caches: Caches, which are compact and quick memory modules placed closer to the CPU than the main memory, may be present in the CPU. Caches reduce memory access latency to boost performance by storing frequently used instructions and data. When the cache is full, the CPU decides which data to evict while still managing cache activities, which include fetching data from memory into the cache.
Coordinating with Other System Components: Through interfaces and buses, the CPU communicates with various system parts, including memory, input/output devices, and peripheral devices. It manages the transmission of information and commands between various parts, guaranteeing proper synchronisation and communication.
Arithmetic and Logic Operations: The CPU performs basic arithmetic operations like addition, subtraction, multiplication, and division. It also handles logical operations such as comparisons, bitwise operations, and boolean operations.
Control Unit: The CPU includes a control unit that coordinates and manages the execution of instructions. It controls the flow of data between the CPU, memory, and other peripherals.
Virtual Memory Management: The CPU works in conjunction with the operating system to manage virtual memory, which allows processes to use more memory than physically available. It handles memory addressing, page table lookups, and swapping data between RAM and disk storage.
Interrupt Handling: The CPU handles interrupts, which are signals from hardware devices or software that require immediate attention. It suspends the current execution, saves the state, and transfers control to the appropriate interrupt handler.
I/O Operations: The CPU communicates with input and output devices, such as keyboards, mice, displays, and storage devices. It coordinates data transfers between these devices and the computer's memory.

Overall, the CPU performs a wide range of functions to ensure the smooth execution of instructions, manipulation of data, and coordination of various components in a computer system.

COMPREHENSION

Answer the following questions:

1. Why is the CPU referred to as the “brain of the computer” according to the text?
2. List the three types of instructions that the CPU interprets and executes.
3. Name all nine elements of a CPU mentioned in the text.
4. What is the role of the Arithmetic Logic Unit (ALU) during instruction execution?
5. What does the control unit do after instructions are fetched?
6. What are registers used for within the CPU?
7. How does the CPU handle an interrupt when it occurs?
8. What is the purpose of caches in relation to memory access latency?
9. The CPU works with the operating system to manage which type of memory?

10. What happens during an I/O operation involving a keyboard or display?

Task 1. Read the following statements and decide if they are True/False or Not Given.

1. The CPU is known as the brain of the computer exclusively because it performs calculations faster than any other component.
2. Instructions for controlling flow are among the three types of instructions that the CPU interprets and executes.
3. The Arithmetic Logic Unit (ALU) is responsible for both arithmetic operations and data transfer between registers and memory.
4. The Control Unit decodes instructions to determine the type of operation and the data involved.
5. Registers temporarily store operands, addresses, and other information needed for calculations.
6. The CPU can only handle interrupts that originate from external hardware devices such as keyboards and timers.
7. Caches are located farther from the CPU than main memory to improve heat dissipation.
8. When the cache is full, the CPU decides which data to evict based on a predefined algorithm.
9. The CPU communicates with memory and peripheral devices exclusively through wireless interfaces.
10. The Control Unit coordinates the flow of data between the CPU, memory, and other peripherals.
11. Virtual memory management allows a process to use exactly the same amount of memory as physically installed in the system.
12. Swapping data between RAM and disk storage during virtual memory management has no effect on processing speed.
13. I/O operations involve coordinating data transfers between input/output devices and the computer's memory.
14. The CPU performs division operations exclusively through the Floating-Point Unit, never through the ALU.
15. The text explicitly states that the pipeline allows multiple instructions to be processed simultaneously at different stages.

SPEAKING AND DISCUSSION

Discuss the following questions with a partner or in small group.

1. The text describes the CPU as carrying out "arithmetic and logical operations." Why would a purely arithmetic CPU (without logical operations) be insufficient for modern computing?
2. The CPU uses a program counter to determine the next instruction. What would happen if the program counter became corrupted due to a hardware fault?
3. Caches store frequently used instructions and data to reduce memory access latency. Why might a larger cache not always lead to better performance despite storing more data?
4. The text states that the CPU handles interrupts from both internal and external events. Why would a keyboard interrupt need to be processed faster than a timer interrupt?
5. Virtual memory management allows processes to use more memory than physically available. What performance penalty might occur when the system relies heavily on swapping data between RAM and disk storage?

6. The CPU coordinates with other system components through buses and interfaces. If the bus speed is significantly slower than the CPU clock speed, where would the bottleneck occur and why?
7. The text mentions that the CPU manages register contents and data transfer between registers and memory. Why are registers made of faster but more expensive memory technology compared to main RAM?
8. The pipeline is listed as an element of the CPU but not explained. Based on the fetch-decode-execute cycle described, why would pipelining improve throughput even though individual instructions take the same amount of time?
9. The Floating-Point Unit (FPU) is listed separately from the ALU. Why would integer arithmetic (ALU) and floating-point arithmetic (FPU) require separate hardware units?
10. The text says the CPU “saves the state” of a paused program when handling an interrupt. What specific information would need to be saved to resume execution correctly?

Read and translate the text.

What is Register in Digital Electronics?[6]

A register is a small and temporary storage unit inside a computer's (CPU). It plays a vital role in holding the data required by the CPU for immediate processing and is made up of flip-flops. It usually holds a limited amount of data ranging from 8 to 64 bits, depending on the processor architecture.

Registers act as intermediate storage for data during arithmetic logic and other processing operations. In this article, we explore the definition, examples, applications, advantages, and disadvantages of registers in computer systems.

What is Register?

A register is a tiny, fast storage memory within the central processing unit (CPU) or the arithmetic logic unit (ALU) of a computer. Registers are utilized for a variety of functions in handling and controlling instructions and data and play an important role in the operation of a computer's CPU.

There are different types of registers listed below :

- Accumulator Register
- Program Counter (PC) Register
- General-Purpose Registers
- Instruction Register (IR)
- Memory Address Register (MAR)
- Memory Data Register (MDR)
- Stack Pointer (SP)
- Floating-Point Registers

Fetch

When we talk about "fetch'' in computers, it is like a computer going on fetching next instruction which has to be followed. The first stage is where computer picks his next operation from its memory - this allows it to know what task has to come at the next moment and provides smooth execution.

Decode

When we talk about "decode'' in computers, computer gets instruction out of memory and decodes it by figuring out what would be the meaning of the instruction that was fetched. After getting the instruction the computer has to decode it so as to know what will be its next course

[6] https://www.geeksforgeeks.org/electronics-engineering/what-is-register-digital-electronics/

of action as per that instruction. This is an important step since it shows how best a computer can perform a certain task.

Execute

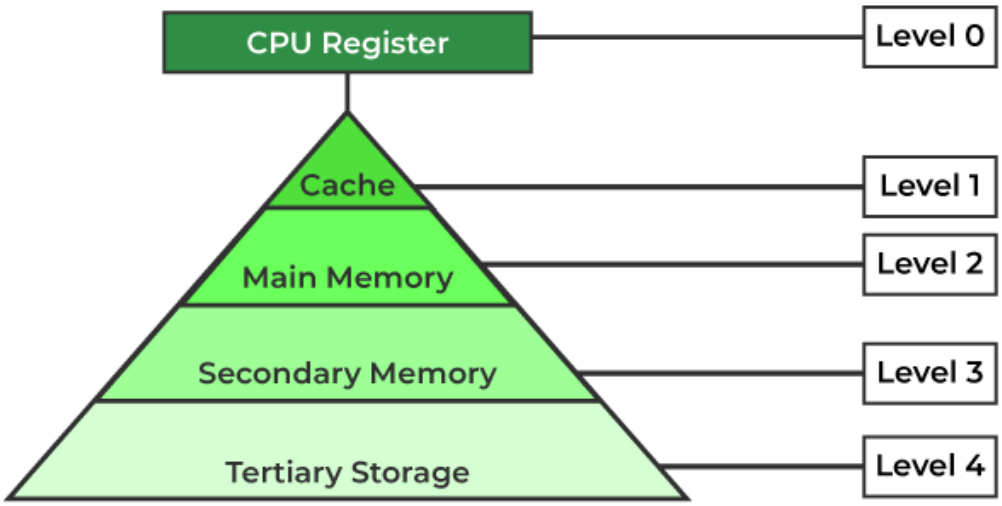
The term "execution" as used in computers refers to an action or task the computer actually undertakes as specified in the instruction that has been fetched and decoded. Once it has understood what is required of it, the computer carries out a certain operation. This could involve performing calculations, transferring data from one location to another, or accomplishing any other task defined by the instructions provided. This is a finishing process marking the end of the instruction cycle after which time it proceeds with the next command so as to maintain its rhythm.

Register Definition

In simple words, we can say that Register is very small, tiny storage unit which is temporary in nature but fast storage memory of a computer.

Memory Hierarchy and the Role of Registers

Computer systems have a memory hierarchy that includes multiple levels of memory with varying access speeds and capacities. At the top of this hierarchy are the CPU registers, which play a vital role in enhancing CPU performance. Registers are small**,** high-speed storage units located within the CPU itself, providing fast access to frequently used data.



Memory Hierarchy

The memory hierarchy typically includes the following levels:

1. **CPU Registers: The** temporary storage units within the CPU are directly accessible by the processor. They hold data required for immediate processing and act as intermediate storage during operations.
2. **Cache Memory: The** fast memory located between the CPU and main memory. Cache memory exploits the principle of temporal locality, caching frequently accessed data to reduce the need to access slower main memory frequently.
3. **logic, (RAM): S**lower than cache memory but larger in capacity the main memory holds data and instructions that are currently being used by the CPU.
4. **Secondary Storage**: The slowest but highest-capacity memory used for the long-term storage of data and programs.

Types of Registers

There are different types of registers listed below as per the different parameters :

- Accumulator Register

- Program Counter (PC) Register
- General-Purpose Registers
- Instruction Register (IR)
- Memory Address Register (MAR)
- Memory Data Register (MDR)
- Stack Pointer (SP)
- Floating-Point Registers
- Index Register (IR)
- Memory Buffer Register (MBR)
- Data Register (DR)

Accumulator Register

The accumulator acts as a central point for arithmetic and logical operations within the CPU. It fetches data from memory and stores intermediate results during calculations. Arithmetic operations such as addition, subtraction, multiplication, and division often take place in the accumulator. The final result may be stored in the accumulator or transferred to other registers or memory locations.

Program Counter (PC) Register

The program counter is a special register that keeps track of the memory address of the next instruction to be fetched and executed. As the CPU executes each instruction in sequence the program counter is updated to indicate the next instruction's address in memory. This process continues until the program's execution is complete.

General-Purpose Registers

General-purpose registers are versatile because they can hold data and memory addresses. They are used for various calculations and data manipulation tasks during program execution. General-purpose registers are essential for performing arithmetic and logical operations on data stored in the CPU.

Instruction Register (IR)

The instruction register holds the currently fetched instruction from memory. It allows the CPU to decode and execute the instruction based on its opcode and operands.

Memory Address Register (MAR)

The memory address register stores the memory address of data or instructions to be accessed or written in memory. It plays a crucial role in memory operations by indicating the location of the data or instruction the CPU needs to access.

Memory Data Register (MDR)

The Memory Data Register holds the actual data fetched from or written to memory. When the CPU retrieves data from memory, it is temporarily stored in the MDR before being processed further.

Stack Pointer (SP)

The stack pointer is used in stack-based memory operations. It keeps track of the top of the stack. which is a region of memory used for temporary storage of data and return addresses during function calls.

Floating-Point Registers

The Floating-point registers are specialized for handling floating-point numbers and performing floating-point arithmetic operations. These registers can store and manipulate floating-point numbers with higher precision.

Index Register (IR)

The Index Register is a kind of useful instrument in the computer that tracks specific locations of data. It is like a small helper that makes it easy for the computer to get to where memory is stored quickly and process it.

Memory Buffer Register (MBR)

The Memory Buffer Register can be viewed as a temporary store in the computer where data takes time off when it is being transferred from one place to another by the CPU. This register can be seen as some kind of traffic police man, ensuring smooth movements of information between CPU and memory without any bumps.

Data Register (DR)

The Data Register is an example of a fast storage place in the computer where information rests while all these processes take place on the same machine. It's a location where transient amounts of data are kept within the system for temporary usage like pit stops within then performing calculations or tasks or others.

Applications of Registers

- **Arithmetic and Logic Operations:** The Registers are extensively used during arithmetic and logic operations in the CPU. They temporarily store operands, intermediate results, and flags facilitating quick and efficient calculations.
- **Instruction Execution:** The Registers play a crucial role in the execution of machine instructions. They hold instructions and data needed for immediate processing ensuring smooth program execution.
- **CPU Pipelining:** The pipelining is employed to increase performance by overlapping the execution of multiple instructions. Registers help in holding intermediate results and data between different stages of the pipeline.
- **Context Switching:** the CPU switches between different processes or threads. it needs to save and restore the context of each process. Registers are used to store the CPU's state including the program counter stack pointer and general-purpose registers during context switching.
- **Memory Addressing:** The memory address register (MAR) and memory data register (MDR) are involved in memory operations. MAR holds the memory address being accessed. while MDR temporarily stores data fetched from or written to memory.
- **Input/Output Operations:** The Registers are used to buffer data during input/output operations. Input data is stored in input registers and output data is temporarily held in output registers before being processed further.
- **Floating-Point Operations:** To Specialized floating-point registers are used for handling floating-point arithmetic operations. These registers can store and manipulate floating-point numbers with higher precision.
- **Control Unit Operations: The** Registers play a critical role in the control unit of the CPU. They hold control signals and flags that determine the control flow and sequencing of the instructions during program execution.

Advantages and Disadvantages of Registers

Advantages and Disadvantages of Registers are given below :

Advantages of Registers

- **Speed:** The Registers offer fast access times due to their proximity to the CPU, enhancing overall system performance.
- **Data Processing Efficiency:** They enable quick data manipulation, reducing the need to access slower main memory frequently.

Disadvantages of Registers

- **Limited Capacity:** The Registers have a small size, restricting the amount of data they can hold at a time.
- **Cost:** The Registers are made from flip-flops and require more hardware, contributing to the overall cost of the processor.

COMPRESSION

Answer the following questions:

1. What term is used to describe a minuscule, transient storage unit located within a computer's central processing unit, and from what fundamental electronic components is it constructed?
2. Apart from the CPU itself, what other major processing unit within a computer is mentioned as containing registers?
3. According to the passage, what is the typical range of bits that a register can hold, and upon what factor does this capacity depend?
4. What are the four distinct levels of the memory hierarchy described in the text, ordered from fastest access to slowest?
5. Which specific register is described as a "central point for arithmetic and logical operations," and what kind of results does it typically store during calculations?
6. What is the primary function of the Program Counter (PC) register, and what happens to its content after each instruction is fetched?
7. The Instruction Register (IR) holds an instruction after it has been fetched. What two components of that instruction does the CPU subsequently use to decode and execute it?
8. What critical role does the Memory Address Register (MAR) play during memory operations, and what does the abbreviation MAR stand for?
9. How does the Memory Data Register (MDR) differ from the MAR in terms of the type of information it holds?
10. In the context of stack-based memory operations, which register keeps track of the top of the stack, and what is this register commonly called?
11. What distinguishes floating-point registers from general-purpose registers in terms of the type of numbers they are designed to handle?
12. According to the passage, what is the function of an Index Register (IR) – noting that this abbreviation shares the same letters as the Instruction Register?
13. The Memory Buffer Register (MBR) is compared to what real-world entity (a "traffic police man"), and what does this analogy illustrate?
14. What is the role of the Data Register (DR) as described in the text, and during what kind of operations is it typically used?
15. Registers are extensively used during arithmetic and logic operations. What three categories of information do they temporarily store to facilitate efficient calculations?
16. In the context of CPU pipelining, how do registers contribute to overlapping the execution of multiple instructions?
17. During a context switch, what specific components of the CPU's state are saved and restored using registers?
18. Which two registers are specifically involved in memory addressing, and what does each one hold during a memory operation?
19. How are registers utilised during input/output operations, and what distinguishes input registers from output registers in this context?
20. According to the final section of the text, what are the two primary advantages and two primary disadvantages of using registers?

Task 1. Read the following statements and decide if they are True/False or Not Given.

1. Not a single register in a computer system is capable of storing more than 64 bits of data at any given moment.

2. The arithmetic logic unit (ALU) is the only component within a computer that does not contain any registers whatsoever.
3. A register is constructed from flip-flops and provides temporary, high-speed storage that is directly accessible by the processor.
4. Cache memory, despite being faster than main memory, still cannot match the access speed of CPU registers.
5. The accumulator register is exclusively used for logical operations and never participates in arithmetic calculations.
6. After the Program Counter fetches an instruction, it is typically decremented so that the CPU can re-execute the same instruction if needed.
7. The Instruction Register holds not only the fetched instruction but also the complete set of all previously executed instructions for debugging purposes.
8. When the CPU decodes an instruction stored in the Instruction Register, it examines both the opcode and any associated operands.
9. The Memory Address Register (MAR) stores the actual data that has been fetched from memory, while the Memory Data Register (MDR) stores the address of that data.
10. In stack-based operations, the Stack Pointer (SP) always points to the bottom of the stack rather than the top.
11. Floating-point registers are incapable of storing integers and can only hold numbers with fractional components.
12. The Index Register (IR) and the Instruction Register (IR) are identical in function despite sharing the same abbreviation.
13. The Memory Buffer Register (MBR) is compared to a traffic police officer because it directs the flow of information without modifying it.
14. The Data Register (DR) serves as a temporary holding location for data while calculations or tasks are performed.
15. Registers are used during arithmetic operations to store operands, intermediate results, and flags – but not the final results of calculations.
16. In CPU pipelining, registers are completely unnecessary because all intermediate results can be passed directly between execution units.
17. During a context switch, the CPU saves only the Program Counter and ignores all general-purpose registers to save time.
18. The Memory Address Register (MAR) and the Memory Data Register (MDR) both participate in every memory read or write operation.
19. Input/output operations never use registers because data from I/O devices is written directly to main memory without any buffering.
20. The two disadvantages of registers mentioned in the text are their high power consumption and their tendency to lose data when power is removed.

SPEAKING AND DISCUSSION

Discuss the following questions with a partner or in a small group.

1. The text states that registers are "tiny, fast storage memory" located within the CPU. Why would placing registers outside the CPU (for example, on the motherboard) significantly reduce their speed advantage?
2. Registers are made of flip-flops, while main memory (RAM) uses different technology. Based on the memory hierarchy diagram, explain why a computer cannot replace all RAM with registers despite registers being much faster.

3. The Program Counter automatically increments after fetching an instruction. What would happen if the PC did not increment and instead remained pointing to the same address? Explain using the concept of sequential execution.
4. The Instruction Register holds the current instruction only temporarily. Why is it necessary to have a separate register for the instruction rather than executing it directly from memory?
5. The text mentions that general-purpose registers can hold both data and memory addresses. What advantage does this flexibility provide over specialised registers that can hold only one type of information?
6. The Memory Address Register (MAR) and Memory Data Register (MDR) work as a pair. Why is it inefficient to combine them into a single register that holds both the address and the data?
7. The stack pointer (SP) keeps track of the top of the stack. If the SP were corrupted (changed to an incorrect value), what kinds of program errors might occur?
8. Floating-point registers are specialised for numbers with decimal points. Why can't general-purpose registers handle floating-point operations with the same efficiency?
9. The text lists the Index Register as a tool that "tracks specific locations of data." How does this differ from the function of the Program Counter, and in what scenario would an index register be more useful than incrementing the PC?
10. The Memory Buffer Register is described as a temporary stop for data moving between CPU and memory. Without the MBR, what additional problem would arise when the CPU needs to read data from memory and perform an operation on it simultaneously?
11. The text states that registers are used during context switching to save the CPU's state. Why is it insufficient to save only the program counter during a context switch?
12. Pipelining overlaps the execution of multiple instructions using registers between pipeline stages. What would happen to pipeline performance if registers were too slow to write and read within one clock cycle?
13. The advantages of registers are speed and data processing efficiency, while disadvantages are limited capacity and cost. Why does increasing register size (e.g., from 32 to 64 bits) not proportionally increase cost as much as increasing RAM capacity?
14. The memory hierarchy places registers at the top, then cache, then RAM, then secondary storage. If a program's working set of data fits entirely in registers, why might it still need to access cache or RAM?
15. The accumulator register acts as a central point for arithmetic operations. In a modern CPU with many general-purpose registers, is an accumulator still necessary? Justify your answer based on the text's description.
16. The text mentions that instructions are "decoded" after being fetched into the Instruction Register. What would the CPU need to do differently if the instruction format were variable-length rather than fixed-length?
17. During input/output operations, input data is stored in input registers. Why is buffering necessary – why can't the CPU process data directly from an I/O device as it arrives?
18. The control unit uses registers to hold control signals and flags. How do flags (e.g., zero flag, carry flag) stored in a register influence conditional branch instructions?
19. The text lists the Memory Buffer Register (MBR) separately from the Memory Data Register (MDR). Are they the same register with different names, or do they serve distinct purposes? Use evidence from the text.
20. Considering both advantages and disadvantages, explain why embedded systems (e.g., a microwave controller) might use fewer and smaller registers than a desktop CPU, even though registers improve speed.

Rendering Assignments

Render the following texts into English.

Двоичный код[7]

Двоичный код (бинарный код) — система записи информации с помощью двух символов — 0 и 1. Он является основой цифровых вычислений, так как компьютеры используют внутреннюю архитектуру и операции, основанные на двоичной системе. Это обеспечивает возможность представления и обработки информации в виде электрических импульсов, с которыми работает современная вычислительная техника. Понимание двоичного кода необходимо для программирования, сетевых технологий и других областей, связанных с цифровой информацией.

Что такое двоичный код

Когда мы говорим о двоичном коде, мы должны сначала разобраться в понятии «бит». Бит (сокращение от binary digit) представляет собой самую маленькую единицу информации. Он может принимать значение либо 0, либо 1. Используя комбинацию битов, можно представлять различные виды информации: числа, символы (буквы, знаки препинания, спецсимволы и т. д.), изображения, звуки и т. д.

Принцип работы современных компьютеров также имеет двоичную природу. Носителем информации в них является электрический заряд или импульс. Например, оперативная память содержит в себе множество ячеек, в которых этот заряд может либо отсутствовать, либо присутствовать, что соответствует 0 и 1. Комбинации таких ячеек с отсутствием или присутствием электрического заряда и являются физическим кодированием информации. Аналогично данные сохраняются на SSD, flash-картах. А вот на магнитных жестких дисках роль мельчайшего физического носителя информации играет ячейка из ферромагнетика, которая также принимает одно из двух состояний — намагниченное или ненамагниченное. Считывая информацию с жесткого диска специальной головкой, компьютер переводит это состояние в соответствующий электрический сигнал.

Иными словами, непосредственно в электронных вычислительных устройствах кодирование информации осуществляется с помощью бинарных пар состояний различных физических сущностей (электрического заряда или импульса, намагничивания и т. д.). А цифровой бинарный код — те самые 0 и 1 — это абстрактная двоичная запись таких состояний.

История двоичного кода

Идея двоичного представления информации появилась в глубокой древности. Одним из первых исторических систем двоичного кода был древний китайский принцип инь-ян. В нем использовались различные комбинации линий, чтобы представить двоичные числа. Эта система применялась в китайской космологии и философии, а также созданных на их основе магических, астрологических и гадательных практиках и не была связана непосредственно с вычислениями.

Применительно к математике и вычислениям идея использования двоичного кода была представлена в 1679 году германским математиком и философом Готфридом Лейбницем. Он предложил применять ее для представления чисел и логических операций. Это стало отправной точкой для развития двоичной арифметики и логики, которые сегодня лежат в

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7 https://blog.skillfactory.ru/glossary/dvoichnyy-kod/

основе работы компьютеров. Но в силу технических ограничений того времени реализовать вычислительные устройства, работающие по такому принципу, было невозможно.

Настоящий прорыв в истории двоичного кода произошел в начале и особенно середине XX века. Это связано с развитием электронных компьютеров и появлением электронных элементов, способных оперировать сигналами в двоичном формате. В 1947 году был создан первый транзистор, который стал основой для разработки электронных компонентов, способных работать с двоичным кодом. В этот период математики Джон Тьюки, Алан Тьюринг и Клод Шеннон были ключевыми фигурами в разработке теории двоичного кода для обработки информации. В частности, Тьюки впервые предложил термин bit, а Шеннон популяризировал его в своей фундаментальной статье «Математическая теория связи».

Единицы компьютерной информации

Выше уже было сказано, что бит — это минимально возможная единица информации, принимающая одно из двух взаимоисключающих значений: да/нет или 1/0. Однако сам по себе практической пользы бит не имеет, потому что его информационная емкость слишком мала, чтобы сохранить или передать хоть какие-то значимые данные.

Блок из двух битов уже может принимать 4 возможных значения: 00, 01, 10, 11. Это уже позволяет закодировать некоторые числа, например 2 и 3. Совокупность трех битов дает уже 8 возможных значений. А блок из 8 битов позволяет составить 256 возможных комбинаций, чего вполне хватает для кодирования букв латинского алфавита со знаками препинания и спецсимволами. Благодаря данной практической ценности именно это количество битов стало считаться минимальной компьютерной единицей информации и получило название байт.

Но на практике кодировать приходится не просто отдельные буквы, а гораздо бóльшие массивы данных: целые тексты, состоящие из тысяч и миллионов символов, аудио-, видеофайлы, изображения, программы и т. д. Для их хранения в двоичном коде необходимы тысячи, миллионы и миллиарды байтов. Чтобы упростить оперирование такими величинами, разработчики компьютерной техники стали оперировать более крупными обозначениями: 1 килобайт — 1024 байта, 1 мегабайт — 1024 килобайта, 1 гигабайт — 1024 мегабайта и т. д.

Как перевести в двоичный код число

Двоичная система исчисления, в отличие от привычной нам десятичной, основана на использовании только двух символов — 0 и 1.

Для перевода числа из десятичной системы в двоичную нужно несколько раз разделить его на 2 и сохранять остатки деления. Для начала возьмем число, которое мы хотим перевести, и поделим его на 2. Запишем остаток от деления (0 или 1) в качестве самого младшего бита (самого правого символа) двоичного числа.

Затем возьмем результат деления и снова поделим его на 2. Сохраняем остаток от деления и добавляем его как следующий бит двоичного числа, помещая его слева от предыдущего бита. Продолжаем выполнять эти шаги, пока не достигнем значения 0. Когда делимое становится равным 0, запись числа в бинарном виде завершается.

Рассмотрим пример для наглядности. Пусть у нас есть десятичное число 17, которое мы хотим перевести в двоичный вид

Шаг 1: 17 / 2 = 8, остаток 1 (записываем 1 как младший бит)
Шаг 2: 8 / 2 = 4, остаток 0 (добавляем 0 слева от предыдущего бита)
Шаг 3: 4 / 2 = 2, остаток 0 (добавляем 0 слева от предыдущего бита)

Шаг 4: 2 / 2 = 1, остаток 0 (добавляем 0 слева от предыдущего бита)
Шаг 5: 1 / 2 = 0, остаток 1 (добавляем 1 слева от предыдущего бита)
Запись битов: 10001
Таким образом, десятичное число 17 в двоичной системе счисления будет представлено как 10001. Перевод с двоичного кода обратно в десятичную систему происходит в обратном порядке.
С помощью двоичной системы можно закодировать не только целые, но и дробные числа. Также она позволяет записывать числа с положительным или отрицательным знаком, для которого специально выделяется отдельный бит информации. В случае чисел с отрицательным знаком производится инвертирование всех битов, позволяющее заменить операции вычитания, в которых участвуют такие числа, операциями сложения. Это необходимо для нормальной работы компьютерного процессора.
Как закодировать в двоичный код слово, изображение, звук и видео
Двоичная кодировка является основной методикой для представления в цифровой форме различных типов данных, таких как тексты, изображения, аудио и видео. Ниже приведены основные аспекты двоичного кодирования для каждого из этих типов данных.
Кодирование текста. Представление символов в двоичном коде происходит следующим образом:

- Каждый символ текста представляется в виде числа с помощью кодировочной таблицы, такой как ASCII или Unicode.
- Число кода символа преобразуется в его двоичное представление. Например, символ A в кодировке ASCII имеет число 65, и его двоичное представление будет 01000001.

Существует множество различных систем перевода текста в бинарный код — например, UTF-8, KOI8, ASCII и т. д. Самым универсальным считается стандарт Unicode, позволяющий кодировать символы практически всех языков мира, а также спецсимволы, арифметические знаки, обозначения денежных единиц и т. д.
Кодирование растровых изображений. Оно происходит по следующему алгоритму:

- Изображение разбивается фиксирующим устройством (цифровой камерой или сканером) на отдельные точки (пиксели), из которых оно состоит.
- У каждого пикселя есть такой параметр, как цвет. Цвет пикселя может разбиваться на базовые составляющие по определенному алгоритму — например, в системе RGB это красный, зеленый и синий.
- Именно количество каждого из них указывается в двоичном коде картинки для каждого пикселя. Также переводятся в двоичный вид такие параметры, как прозрачность, интенсивность цвета и яркость.

Чем больше битов используется для кодирования пикселя, тем больше возможностей для смешивания трех каналов и тем более насыщенными будут цвета в изображении.
Кодирование аудио. Перевод звуковой информации в двоичный код происходит примерно схожим образом, только звук фиксирующим устройством (например, микрофоном) переводится в электрические колебания. Затем звуковая карта компьютера или звукозаписывающей аппаратуры разбивает (дискретизирует) эту волну на отдельные временные участки. У каждого такого участка будет своя интенсивность, характеризующая отклонение от начального значения. Именно числовое значение интенсивности каждого фрагмента записывается в двоичный код.
Кодирование видео. Особенность этого процесса состоит в том, что он подразумевает одновременное кодирование звука и графического изображения. Первое происходит по описанному выше алгоритму. Визуальная же составляющая кодируется покадрово. Причем происходить это может двумя путями:

- Полностью кодируется каждый кадр, представляющий собой отдельное растровое изображение. Этот способ обеспечивает максимальную точность, но увеличивает размер конечного файла.
- Полностью кодируется только первый кадр, а дальше в двоичный код переводятся только изменения изображения в последующих кадрах. Это сокращает размер файла, но может несколько снижать качество.

Однако с учетом того, что видео дискретизируется с частотой 24 и более кадров в секунду, снижение точности при использовании второго способа минимально. Поэтому на практике такой алгоритм используется чаще.

Преимущества двоичного кода

Простота и надежность. Двоичный код состоит только из двух состояний — 0 и 1, что делает его очень простым в использовании и интерпретации. Компьютеры и электронные устройства обрабатывают и распознают его очень эффективно. Также двоичный код устойчив к помехам и ошибкам при передаче данных, поскольку каждый бит представлен одним из двух состояний.

Легкость реализации. Множество электронных устройств, включая компьютеры, используют двоичную систему для работы, поскольку она является базовой и наиболее удобной для электронных компонентов. Они могут легко обрабатывать и хранить двоичные данные, что делает их эффективными в использовании ресурсов системы.

Универсальность. Бинарное кодирование является универсальным языком для выражения информации в компьютерном мире. Оно используется для представления текста, чисел и других форматов данных. Будучи стандартным языком обмена информацией, двоичный код облегчает взаимодействие между различными компьютерными системами.

Масштабируемость. Двоичный код обеспечивает большую гибкость и масштабируемость в представлении и обработке данных. Путем добавления большего количества битов можно повысить точность представления информации или увеличить ее объем. Это особенно важно в области вычислений и обработки сигналов, где точность и детализация данных играют важную роль.

Совместимость. Благодаря широкому использованию двоичного кода, его возможностям и простоте практически все компьютерные системы и электронные устройства обладают его поддержкой. Это обеспечивает совместимость между различными системами и позволяет обмениваться информацией без проблем.

Недостатки двоичного кода

Сложность чтения. Записанная с помощью 0 и 1 информация нечитаема для человека, не обладающего специальными знаниями и обучения. Расшифровка двоичного кода на понятный язык может быть очень сложной и требует использования специальных методов и программ.

Высокая сложность программирования. Написание программы на двоичном коде невозможно без глубокого понимания и экспертизы в области низкоуровневого программирования. Это требует большого количества усилий и времени, особенно для составления длинных и сложных программ.

Ограниченная емкость. Двоичное представление требует больше символов для тех же данных по сравнению с другими системами кодирования. Это может привести к увеличению размера файла или затратам на дополнительное хранение.

Сложность визуализации. Двоичный код трудно визуализировать, особенно для больших наборов данных. Это может затруднять его отслеживание и анализ без использования специальных инструментов и программ.

Таким образом, двоичное кодирование информации остается на данный момент основным способом, используемым в вычислительных устройствах. На нем основана вся логическая архитектура смартфонов, планшетов, ПК, огромных суперкомпьютеров, бортовых вычислительных систем самолетов, космических аппаратов, кораблей и т. д. Тем не менее у него есть недостатки, которые ограничивают развитие компьютерной техники. Возможно, в связи с экспоненциальным ростом потребности в эффективной обработке информации в будущем бинарный код уступит место альтернативным способам кодировки.

Чипы[8]

Интегральная схема, чип — микроэлектронное изделие, имеющее высокую плотность упаковки электрически соединенных элементов и рассматриваемое как единое конструктивное целое. Интегральные схемы делятся на категории по числу элементов. Очевидно, что чем больше элементов размещено в данной области, тем меньше они должны быть. При самых плотных уровнях интеграции элементы схемы могут иметь размеры, сравнимые с размерами атомов. Транзисторные компьютеры, которые были в сотни раз меньше ламповых машин такой же производительности, появились во второй половине 1950-х годов. Единственная часть компьютера, в которой транзисторы не смогли заменить электронные лампы, — это блоки памяти, где вместо ламп стали употреблять изобретенные к тому времени схемы памяти на магнитных сердечниках. Каждый транзистор тогда изготавливался отдельно, а при сборке схем их соединяли и припаивали вручную. В 1958 году американский инженер Джек Килби придумал, как на одной пластине полупроводника получить несколько транзисторов [3—6]. После окончания Иллинойсского университета Килби 10 лет работал с транзисторами в фирме, выпускавшей детали для радиоприемников и телевизоров. В мае 1958 года ему удалось получить место в компании Texas Instruments, выполнявшей в то время военный заказ, связанный с “миниатюризацией” электронных схем. Разработанный тогда метод микромодулей заключался в том, что электронные элементы “печатались” на тончайших керамических пластинках, которые затем спаивались в виде стопки для получения нужной схемы. Килби решил, что данный способ является слишком замысловатым, и стал искать ему замену. Летом, когда сотрудники компании отправились в отпуск, Килби как новичок, еще не заслуживший на это права, остался в лаборатории. Он продолжал свой поиск и вскоре пришел к выводу: можно не только делать сопротивления и конденсаторы из Джек Килби того же полупроводникового материала, что и транзисторы, но и изготавливать все компоненты одновременно, на одной полупроводниковой пластине [3]. Через некоторое время Килби сделал первый грубый опытный образец. Для самой первой интегральной схемы (ИС) была использована тонкая германиевая пластинка длиной 1 см. [Позже интегральные схемы стали называть еще чипами (*chip* — щепка).] Приспособление не отличалось изяществом. Маленькие проволочки, соединяющие компоненты друг с другом и источником питания, просто припаивались. Вся конструкция скреплялась воском. Но схема работала. Фирма объявила о рождении нового устройства в январе 1959 года. А чтобы продемонстрировать потенциальные возможности новой технологии, она построила специализированный компьютер, состоявший из нескольких сотен интегральных схем, объем которого был в 150 раз меньше, чем у подобной машины старого образца. Однако предложенное Килби устройство имело существенные недостатки. Спустя некоторое время Роберт Нойс, будущий основатель фирмы Intel

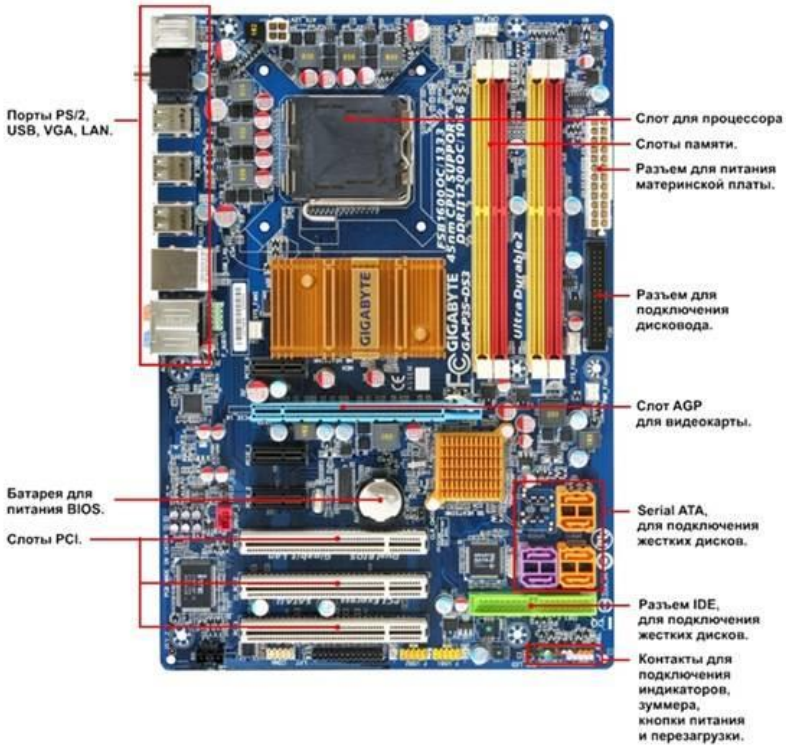
[8] https://informat444.narod.ru/museum/pres/inform/21.htm

(от **inte**grated **el**ectronics — интегральная электроника), нашел более совершенный метод, позволивший помещать на одной пластине и все элементы, и все необходимые соединения между ними. Интегральные схемы дали возможность значительно сократить размеры изделий, избавили от необходимости трудоемкой пайки соединений между элементами, а уменьшение количества соединений способствовало повышению надежности приборов. Весьма важно и то, что повысилась скорость их работы, так как электрические импульсы преодолевали теперь значительно меньшие расстояния. Все это коснулось, конечно, и компьютерной техники: интегральные схемы позволили строить менее громоздкие и более мощные компьютеры. В данную область интегральные микросхемы проникли уже в середине 1960-х годов, заменив сначала логические схемы, состоящие из отдельных транзисторов, а затем и память на ферритовых сердечниках. В 1970 году фирма Intel начала продавать интегральные схемы памяти, и примерно в то же время был сделан еще один очень важный шаг — Маршиан Эдвард Хофф, пришедший в фирму вскоре после ее образования, построил интегральную схему, аналогичную по своим функциям центральному процессору большой ЭВМ — “интегральное микропрограммируемое вычислительное устройство”. Так появился первый микропроцессор (*термин микропроцессор стал употребляться с 1972 года*), получивший позже название Intel-4004.

В 2000 году Джеку Килби (а также Герберту Кремеру и российскому исследователю Жоресу Ивановичу Алферову) была присуждена Нобелевская премия по физике.

Назначение, основные элементы и устройство материнской платы[9]

Материнская плата–это главная монтажная деталь, к которой подключаются внутренние компоненты и внешние, периферийные устройства. Она является базовым элементом любого настольного, переносного, планшетного или карманного компьютера. В этой публикации будет рассмотрено устройство материнской платы, а также назначение ее основных компонентов.



[9] https://systech.ru/vopros-otvet/elementy-i-ustrojstvo-materinskoj-platy

Материнская плата обеспечивает взаимодействие различных устройств ПК. Обязательными ее элементами являются разъем или socket процессора (ЦП), чип-сет, коннекторы для модулей памяти. Кроме этого, обязательными элементами ПК являются: микросхема BIOS, контролеры, различные разъемы для подачи питания, коннекторы подключения элементов и устройств компьютера.

Микросхема BIOS

BIOS представляет собой микросхему, с определенными тестовыми и загрузочными программами для ЦП, ведь на начальном этапе процессор еще не знает, как работать со всеми устройствами, расположенными в ПК. Находится эта микросхема в непосредственной близости от ЦП и запитана от отдельной батарейки — «таблетки», расположенной на плате, рядом со слотами расширения.

Первоначальное тестирование происходит без участия процессора. Только после проверки электронных элементов компьютера на предмет отсутствия короткого замыкания, и соответствия токов в БП, BIOS «разрешает» подавать блоку питания необходимый ток на плату и процессор.

После, ЦП считывает с БИОС определенный алгоритм действий, благодаря которым происходит тестирование работоспособности всех компонентов ПК. Если все в порядке, то производится запуск операционной системы, расположенной на «винчестере». После этого, все управление ПК передается исключительно операционной системе.

Благодаря своей работе, BIOS незаметно для пользователя выполняет одну из важнейших функций в компьютере.

Разъем центрального процессора

Разъем или socket процессора, на материнской плате самый большой. Модели разъемов ЦП различаются по внешнему виду, расположению и числу контактов. В зависимости от модели процессора, они бывают двух типов:

- Сокетный, от анг. «гнездо». Он представляет собой прямоугольный или квадратный коннектор с множеством отверстий-контактов, расположенным по его периметру. Монтируется процессор в такой разъем – горизонтально.
- Слотовый разъем, представляет собой длинный ряд контактов, расположенных в пластмассовом корпусе. Он предназначен для вертикального монтажа ЦП. На сегодняшний день, такой тип разъемов практически не используется.

Назначение и конструктивные особенности чипсета

Чипсет материнской платы обычно представляет собой две разные микросхемы, именуемые «северным» и «южным» мостом. Свои названия они получили исключительно из расположения: северный мост находится возле ЦП, а южный – ниже, возле разъема видеокарты и слотов расширения. Они являются важнейшим связующим звеном между аппаратной частью и процессором ПК. В некоторых современных моделях системных плат может использоваться чипсет, состоящий из одной микросхемы. Это связано с конструктивными особенностями процессора, взявшего на себя функции южного моста.

Северный мост осуществляет связь ЦП с наиболее быстрыми компонентами ПК и с южным мостом. Из-за высокой нагрузки, мост достаточно сильно нагревается, поэтому, чаще всего, его оснащают радиатором для лучшего охлаждения.

Северный мост связан с ЦП высокоскоростной шиной. Наиболее современной, на сегодняшний день, считается шина QPI, от компании Intel.

Для связи моста с видеоадаптером, может использоваться шина AGP 3.0, или PCI Express 3.0, массовый выпуск которой начался с 2012 г.

Микросхемы мостов между собой по одному из представленных интерфейсов: PCI, Hub Link, DMI или HyperTransport.
Южная часть чипсета, согласовывает работу более медленных компонентов ПК, а также обеспечивает передачу данных с этих компонентов на северный чип.
Южный мост состоит из контроллеров: для связи с северной частью чипсета, платами расширения, периферийными устройствами и компонентами, жесткими дисками и другими медленными устройствами
Наиболее распространенными интерфейсами, в современных ПК считаются: USB 2.0 — 3.0; WI-FI и Ethernet. Для связи южного моста с жесткими дисками используются шины PCI, а также более современные шины с SATA и SCSI и SAS. Для согласования работы моста с BIOS чаще всего используется шина LPC с тактовой частотой 33,3 МГц.

Коннекторы модулей памяти и других плат расширения
В любом компьютере существует временное хранилище информации, именуемое оперативной памятью. Она представляет собой модуль с несколькими микросхемами на борту, который монтируется в специальные разъемы, так называемые слоты. Как правило, они расположены рядом с процессором и северной частью чипсета. Слоты модулей памяти оснащены замками, которые изготовлены с защитой от неправильного монтажа. Количество их варьируется от 2-х до 6,в зависимости от стоимости «материнки» и ее форм-фактора.

Важными компонентом материнской платы является слот PCI-Express, для подключения дискретного видеоадаптера, без которого невозможен вывод изображения на монитор (если только в компьютере нет интегрированной видеокарты).
Кроме вышеперечисленных слотов, на плате расположены штыревые разъемы для подключения шин передачи данных жестких дисков. Раньше, для этого использовался интерфейс IDE. На современных ПК, чаще всего используется интерфейс SATA.
На любой «материнке» в обязательном порядке есть порты, для подключения дополнительных периферийных устройств. Как правило, основной набор включает в себя: разъемы питания вентиляторов и кнопок, расположенных на панели системного блока; светодиодов, сигнализирующих о работе «винчестера» и наличии питания. Кроме того, на задней части платы находится: несколько портов USB; разъем LAN для подключения кабеля к сетевой карте; коннекторы аудиовхода и выхода; разъем HDMI и пр. В большинстве моделей можно встретить разъемы PS/2для подключения клавиатуры и манипулятора.
Для питания материнской платы, а также всех электронных и механических компонентов, на ней установлен 20 или 24 контактный разъем, на который приходит напряжение от блока питания, расположенного в корпусе ПК.

Типы системной платы
Форм фактор материнской платы – это определенный стандарт, благодаря которому определяется размер системной платы, ее способ крепления в корпусе ПК, наличие и расположение портов и интерфейсов шин, а также разъемов ЦП и оперативной памяти.

В настоящее время существует несколько форм-факторов системных плат:

- ATX – самая большая по величине материнская плата, которая имеет много слотов и удобное расположение портов вода — вывода.
- EATX – это системная плата ничем не отличающаяся от ATX, кроме размеров, которые составляют 30,5 см. х33 см.

- MicroATX – это уменьшенный вариант материнской платы ATX. Она была разработана для ПК, которым не требуется изменение конфигурации, поэтому она имеет 4 слота расширения. Как правило, такие платы оснащены 24 пиновым разъемом питания и имеют размеры, 24, 5 см. х 24,5 см.
- BTX – материнская плата, разработанная для создания небольших системных блоков. Тем не менее, она имеет 7 слотов расширения при размерах 26.7 х 32.5 см.
- Micro BTX является уменьшенной копией «материнки» BTX, которая имеет 4 слота расширения и размеры 26.7 х 26.4 см.
- MiniITX – самая маленькая, из современных системных плат для компьютера. Ее размеры составляют 17 см на 17 см.
- SSIEEB и SSICEB. Материнские платы этих форм-факторов применяются для создания серверов. Размеры составляют: 30.5 х 33.0 см. и соответственно 30.5 х 25.9 см.

Что такое тактовая частота процессора, на что она влияет, какая должна быть, в чем измеряется, за что отвечает и как ее увеличить[10]

Центральный процессор (ЦП) часто считают «мозгом» компьютера, поскольку он обрабатывает инструкции, поступающие от аппаратного, программного обеспечения и периферийных устройств компьютера. Именно в нем происходит большинство вычислений. С точки зрения вычислительной мощности центральный процессор — самый важный элемент цифровой компьютерной системы. Одной из его важнейших характеристик является тактовая частота. Рассказываем простыми словами, что это такое.
Что такое тактовая частота процессора

Тактовая частота процессора — это частота, на которой он работает, опосредованно выражающая число операций в секунду. По сути, это показатель того, сколько циклов (или инструкций) процессор может выполнить за одну секунду. Каждый цикл позволяет процессору выполнять различные задачи, такие как выполнение инструкции, доступ к памяти или управление операциями ввода/вывода.
Более высокая тактовая частота означает, что процессор может выполнить больше циклов в секунду, что потенциально приводит к ускорению выполнения задач и повышению общей производительности. Однако эффективность тактовой частоты процессора зависит и от других факторов, таких как архитектура процессора, количество ядер и то, насколько эффективно программное обеспечение использует доступные аппаратные ресурсы.
Например, процессор с тактовой частотой *3* ГГц может выполнять *3* миллиарда циклов в секунду. Важно отметить, что хотя тактовая частота может быть показателем производительности, она не является единственным определяющим фактором. Развитие технологий позволило усовершенствовать другие области, такие как многопоточность и энергоэффективность, которые также существенно влияют на производительность компьютера.
Тактовая частота процессора влияет на несколько аспектов производительности компьютера, в том числе на показатели, указанные ниже.
Скорость выполнения инструкций. Более высокая тактовая частота позволяет процессору совершать больше циклов в секунду, а значит, быстрее выполнять инструкции. Это очень важно для задач, которые в значительной степени зависят от однопоточной

[10] https://digitalocean.ru/n/chto-takoe-taktovaya-chastota-processora

производительности, например для некоторых типов вычислений или запуска старого программного обеспечения, предназначенного для одноядерных процессоров.

Общая производительность системы. Тактовая частота процессора влияет на общую скорость работы компьютера. Сюда входит запуск приложений, выполнение сложных программных алгоритмов, обработка данных и выполнение задач по обслуживанию системы. Более быстрый процессор может в определенной степени улучшить отзывчивость и скорость работы компьютерной системы.

Многозадачность. Хотя многозадачность напрямую зависит от количества ядер и потоков процессора, тактовая частота может играть значительную роль в том, насколько гладко система может справляться с несколькими задачами одновременно. Более высокая тактовая частота позволяет повысить производительность каждого ядра, что способствует более плавной работе при одновременном запуске нескольких приложений.

Игры и графика. Многие видеоигры и графические приложения работают как с процессором, так и с графическим процессором. Более высокая тактовая частота процессора может повысить производительность игр, особенно в сценариях, где игра зависит от процессора для расчета физики, искусственного интеллекта и других элементов, влияющих на игровой процесс.

Создание и редактирование контента. Такие задачи, как редактирование видео, *3D*-рендеринг и компиляция больших программ, могут выиграть от более высокой тактовой частоты процессора. Эти приложения часто требуют быстрой обработки сложных вычислений и манипулирования данными, и более высокая тактовая частота может значительно сократить время обработки.

Потребление энергии и тепловыделение. Более высокие тактовые частоты обычно приводят к увеличению энергопотребления и тепловыделения. Это означает, что процессоры с более высокими тактовыми частотами нуждаются в более надежных системах охлаждения для поддержания оптимальной рабочей температуры и предотвращения перегрева, который может привести к снижению производительности или повреждению компонентов.

Важно отметить, что хотя тактовая частота является одним из важнейших факторов производительности процессора, она не единственная. Эффективность архитектуры процессора, количество ядер и потоков, объем кэша и конкретная выполняемая рабочая нагрузка также играют важную роль в определении общей производительности. В современных вычислениях для достижения оптимальной производительности необходим сбалансированный подход, учитывающий все эти факторы.

Понятие «лучшей» тактовой частоты процессора не так однозначно, как может показаться, поскольку оптимальная тактовая частота процессора зависит от различных факторов, включая конкретные задачи, которые вы хотите выполнять, общую архитектуру процессора, а также баланс между производительностью и энергопотреблением.

Если вы используете компьютер в основном для выполнения базовых задач, таких как просмотр веб-страниц, работа с текстом и потоковое видео, вам будет достаточно процессора со скромной тактовой частотой. Для более требовательных задач, таких как игры, редактирование видео, *3D*-рендеринг или научные вычисления, более высокая тактовая частота может повысить производительность, особенно в сочетании с несколькими ядрами, которые эффективно справляются с параллельной обработкой данных.

Эффективность архитектуры процессора (способ обработки инструкций, размер кэша, количество ядер и т. д.) может существенно влиять на производительность. Иногда процессор с меньшей тактовой частотой, но более эффективной архитектурой может превзойти процессор с большей тактовой частотой.

Более высокие тактовые частоты часто приводят к увеличению энергопотребления и выделению тепла. Эффективные системы охлаждения необходимы для поддержания оптимальной производительности и предотвращения перегрева, особенно в ноутбуках и компактных устройствах. Для устройств с батарейным питанием баланс между производительностью и энергоэффективностью имеет решающее значение для обеспечения максимального времени автономной работы.

Современные процессоры часто оснащаются такими технологиями, как турборежим или динамический разгон, которые временно увеличивают тактовую частоту при определенных условиях для повышения производительности. Эти функции позволяют процессору работать на более высокой скорости, когда это необходимо, и сохранять более низкую скорость (и, соответственно, меньшее энергопотребление) при выполнении менее требовательных задач.

На современном рынке для общих вычислений очень распространены процессоры в диапазоне от *2* ГГц до *4* ГГц, которые обеспечивают хороший баланс производительности и энергоэффективности для большинства пользователей. Для высококлассных игр, видеомонтажа и профессиональных нагрузок процессоры могут иметь базовую тактовую частоту в нижней части этого диапазона, но под нагрузкой могут повышаться до более высоких частот (*4* ГГц и выше).

В конечном счете, выбор оптимальной тактовой частоты процессора зависит от ваших конкретных потребностей и того, как вы соизмеряете их с другими факторами, такими как бюджет, энергопотребление и типы приложений, которые вы чаще всего запускаете. При выборе процессора стоит обратить внимание на комплексные бенчмарки производительности, моделирующие реальные задачи, а не выбирать только по тактовой частоте.

Как увеличить тактовую частоту процессора

Увеличение тактовой частоты процессора, известное как «разгон», — это метод, позволяющий заставить процессор работать на более высокой скорости, чем установлено по умолчанию. Этот процесс потенциально может привести к повышению производительности, особенно в таких требовательных задачах, как игры, редактирование видео и *3D*-рендеринг. Однако к разгону следует подходить с осторожностью, поскольку при неправильном подходе он может привести к увеличению энергопотребления, выделению дополнительного тепла и потенциальному сокращению срока службы процессора.

Убедитесь, что ваш процессор разблокирован и может быть разогнан. Процессоры *AMD* обычно позволяют разгонять большую часть своей линейки, а *Intel* обозначает свои разгоняемые процессоры буквой «*K*» или «*X*» в конце номера модели.

Убедитесь, что ваша материнская плата поддерживает разгон. Обычно это указано в спецификациях материнской платы.

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